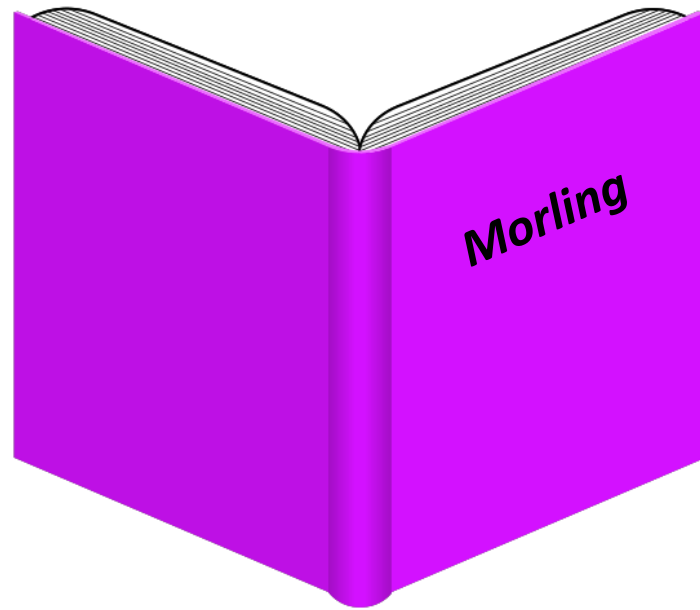
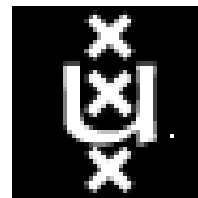




Research Methods and Statistics

Lecture 9: Probability Distributions

Riet van Bork



Clash of books?

- Last year, observer effect, observer bias, and reactivity (terms used in the RMS book) differed from similar concepts in the introductory psychology book.
- In RMS, there is a distinction between observer effect and observer bias:
 - Observer effect (or 'expectancy effect'): the observer influences the subject's behavior
 - Observer bias: the observer's expectation influences their *interpretation* of behavior
- New books, new editions, are the concepts consistent across courses?
- In case there are inconsistencies, let me know. And for the RMS exam, base your answers on the RMS reading material

Today

- **Random variables**
- Probability distributions
 - Discrete variables
 - Three different probability calculations
 - Mean and variability
 - Continuous variables

Two specific distributions:

- Normal distribution (possible for continuous variables)
 - Mean and variability
 - Three different probability calculations
- Binomial distribution (possible for discrete variables)
 - General formulae
 - Mean and variability

Random variables

Random variable: A numerical measure of the outcome of a random phenomenon

“Effectiveness of therapy depends on number of close relationships”

Variable: “effective”: 0 (no), 1 (yes)

Variable: “number of close relationships”: 1, 2, 3,.. Etc (positive integer)

Random variables

Random variable: A numerical measure of the outcome of a random phenomenon

Often, randomness results from the use of random sampling to gather the data

Random phenomena:

Randomly drawing a patient and evaluating whether the treatment is effective

→ Variable “effective”: 0 (no), 1 (yes)

Randomly drawing a student and taking their height

→ Variable “height”: number

It is called a variable because the values vary over something (in this case over people)

Randomly drawing a student and asking their hobby

→ Variable “hobby”: 0 (“tennis”), 1 (“shopping”), 2 (“netflix”), etc.

Randomly drawing a student and counting their number of siblings

→ Variable “number of siblings”: positive integer

Random variables

Random variable: A numerical measure of the outcome of a random phenomenon

Random phenomena:

Randomly drawing a patient and seeing whether the treatment is effective

→ Variable “effective”: 0 (no), 1 (yes)

Categorical

Randomly drawing a student and taking their height

→ Variable “height”: number

Continuous

Randomly drawing a student and asking their hobby

→ Variable “hobby”: 0 (“tennis”), 1 (“shopping”), 2 (“netflix”), etc.

Categorical

Randomly drawing a student and counting their number of siblings

→ Variable “number of siblings”: positive integer

Discrete

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Example

Nine key symptoms (“A” criteria) of Major Depression according to DSM IV:

1. depressed mood
2. loss of interest
3. appetite or weight change
4. Insomnia or hypersomnia
5. psychomotor agitation or retardation
6. fatigue
7. feelings of worthlessness or guilt
8. diminished ability to think or concentrate or indecisiveness
9. thoughts of death or suicide

Example

- If we select a random person living in the Netherlands. How many of these symptoms will that person have?
- → Number of symptoms is a **discrete** random variable
- Possible values: 0, 1, 2, 3, ..., 9
- But not all values are equally probable
 - Probability of “9 symptoms” is much smaller than probability of “0 symptoms”



Image source: pixabay.com



- X = number of symptoms
- Values: $x = 0, x = 1, x = 2, x = 3, x = 4, x = 5, x = 6, x = 7, x = 8, x = 9$

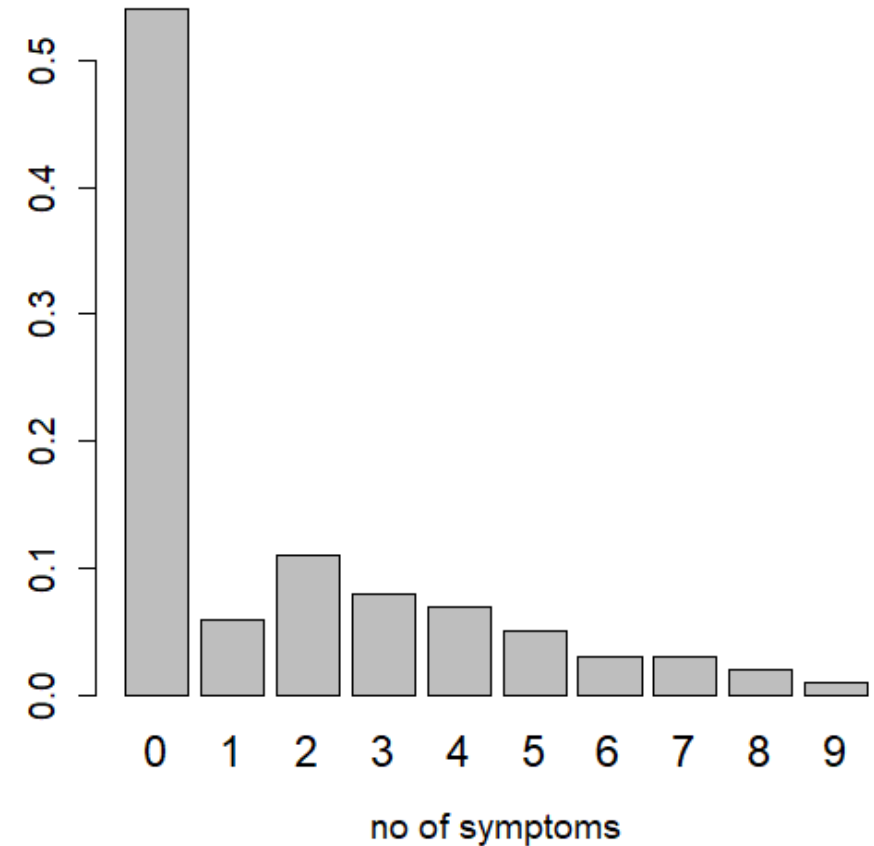
x	$P(x)$
0	0.54
1	0.06
2	0.11
3	0.08
4	0.07
5	0.05
6	0.03
7	0.03
8	0.02
9	0.01



Probability distribution

Sum of these probabilities?

*Total probability is always 1!!
(i.e., $0.54 + 0.06 + \dots + 0.01 = 1$)*



Probability distribution: An overview of how probable each value of a random variable is

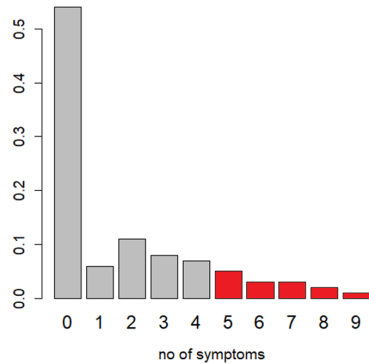
Today

- Random variables
- Probability distributions
 - Discrete variables
 - **Three different probability calculations**
 - Mean and variability
 - Continuous variables

Two specific distributions:

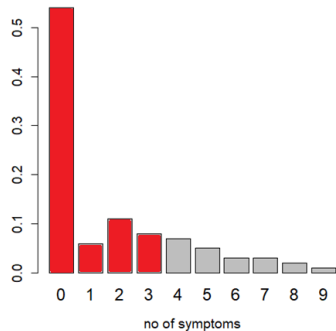
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Three different probability calculations



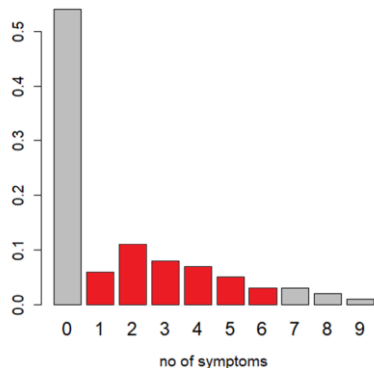
1) Probability of 5 or more symptoms

- $P(X \geq 5)$



2) Probability of 3 or fewer symptoms

- $P(X \leq 3)$

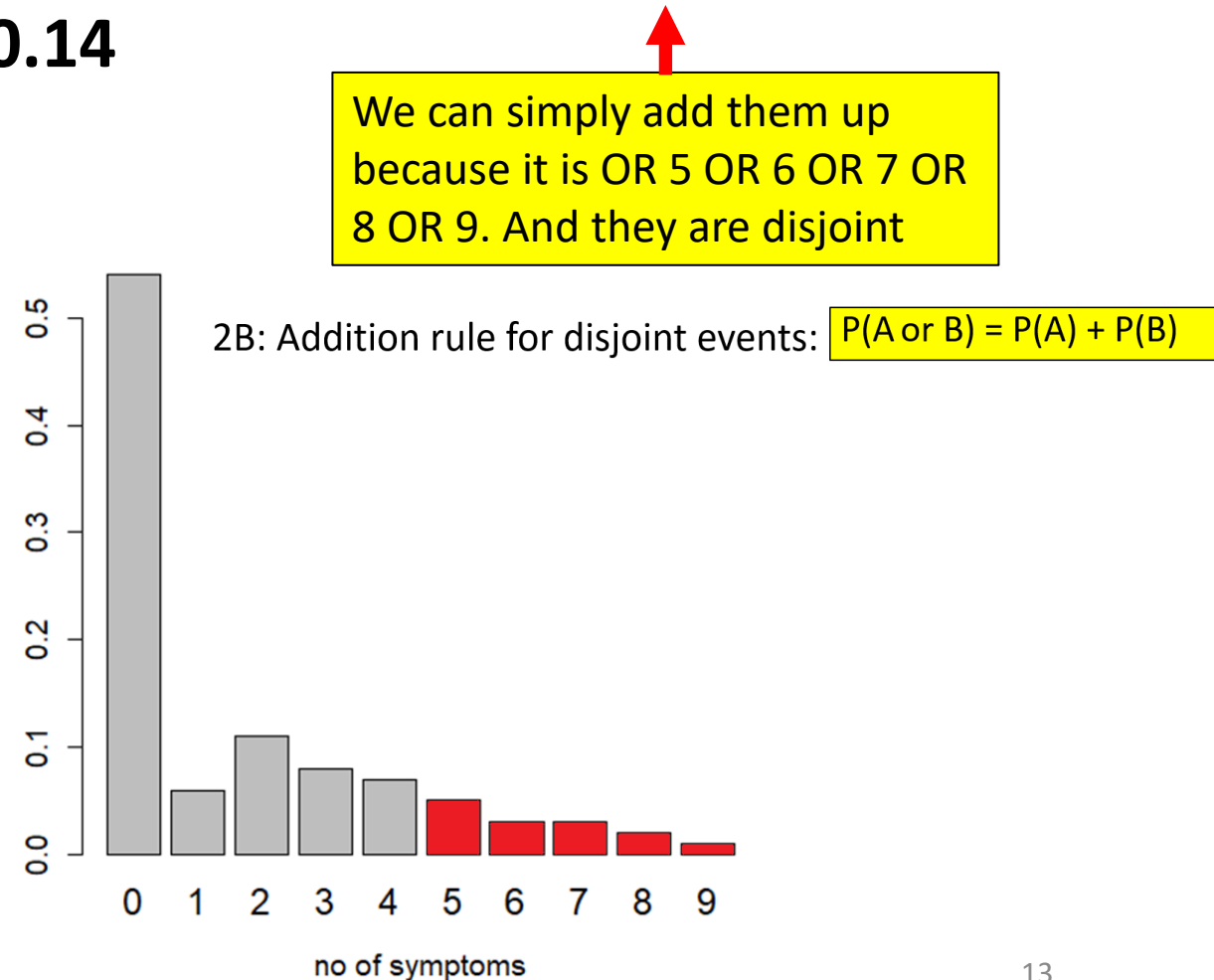


3) Probability of at least 1 and at most 6 symptoms

- $P(1 \leq X \leq 6)$

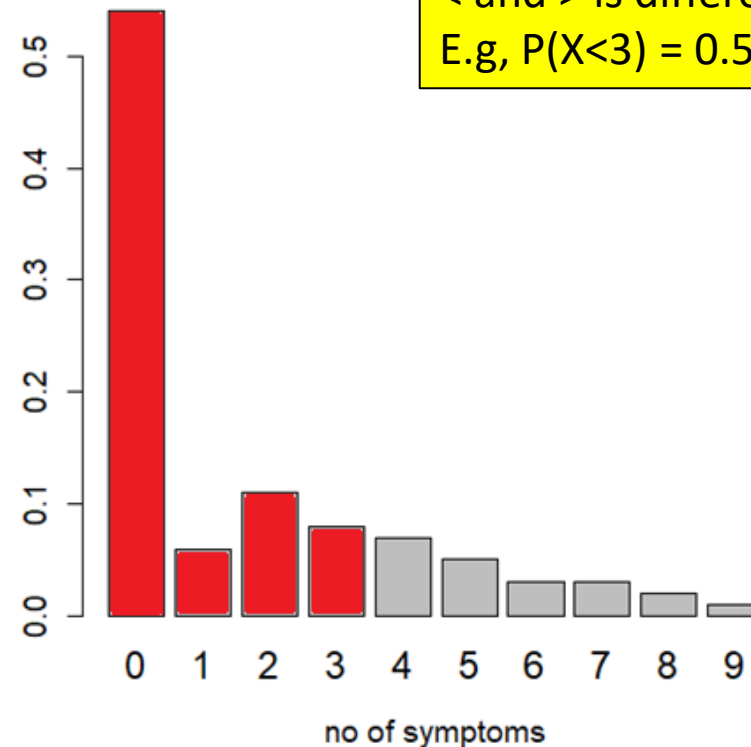
- You are diagnosed as “depressed” if you have 5 or more symptoms
- X = number of symptoms
- $\rightarrow P(\text{depression}) = P(5 \text{ or more}) = P(X \geq 5) = \mathbf{P(5) + P(6) + P(7) + P(8) + P(9)}$
 $= \mathbf{0.05 + 0.03 + 0.03 + 0.02 + 0.01 = 0.14}$

x	$P(x)$
0	0.54
1	0.06
2	0.11
3	0.08
4	0.07
5	0.05
6	0.03
7	0.03
8	0.02
9	0.01



- What is the probability of 3 or fewer symptoms
- X = number of symptoms
- $\rightarrow P(\text{depression}) = P(3 \text{ or fewer}) = P(X \leq 3) = \mathbf{P(0) + P(1) + P(2) + P(3)}$
 $= \mathbf{0.54 + 0.06 + 0.11 + 0.08 = 0.79}$

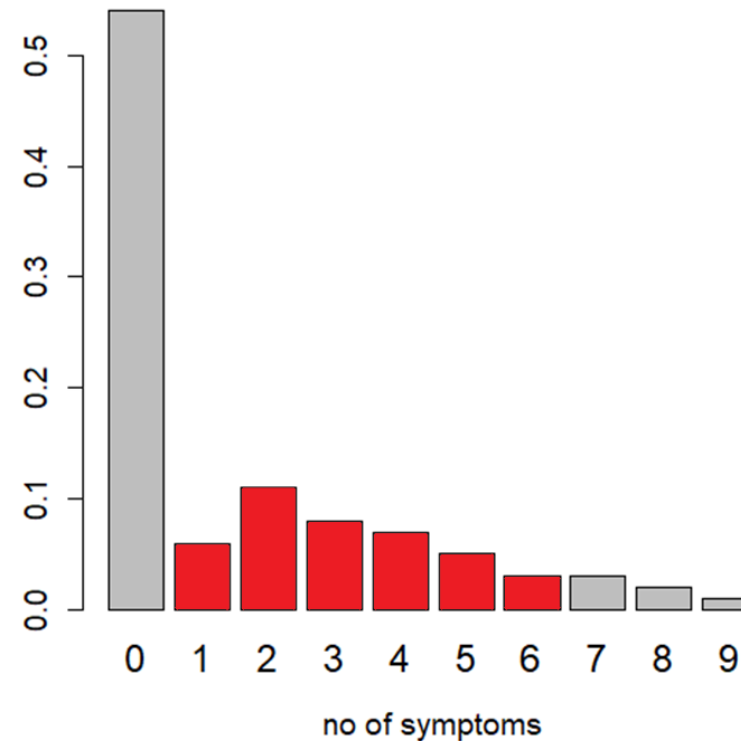
x	$P(x)$
0	0.54
1	0.06
2	0.11
3	0.08
4	0.07
5	0.05
6	0.03
7	0.03
8	0.02
9	0.01



Be careful as for discrete distributions
 $<$ and $>$ is different from $\leq$ and $\geq$
 E.g, $P(X < 3) = 0.54 + 0.06 + 0.11 = 0.71$

- What is probability of number of symptoms between 1 and 6?
- X = number of symptoms
- $\rightarrow P(X=1 \text{ OR } X=2 \dots \text{ OR } X=6) = P(1 \leq X \leq 6) = P(1) + P(2) + P(3) + \dots + P(6)$
- $\rightarrow P(1 \leq X \leq 6) = 0.06 + 0.11 + 0.08 + 0.07 + 0.05 + 0.03 = 0.4$

x	$P(x)$
0	0.54
1	0.06
2	0.11
3	0.08
4	0.07
5	0.05
6	0.03
7	0.03
8	0.02
9	0.01



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Mean (or expected value), μ

$$\mu = \sum xP(x)$$

- a weighted average
- **don't** do: $(0+1+2+3+4+5+6+7+8+9)/10 = 4.5$

Greek letters
refer to
parameters in
the population.
Contrast to $\bar{x}$
(Ch 2, p.96)

x	P(x)
0	0.54
1	0.06
2	0.11
3	0.08
4	0.07
5	0.05
6	0.03
7	0.03
8	0.02
9	0.01

$\mu =$

$$\begin{aligned} & 0 \times P(0) + 1 \times P(1) + 2 \times P(2) + 3 \times P(3) + 4 \times P(4) + 5 \times P(5) + 6 \times P(6) + 7 \times P(7) + 8 \times P(8) + 9 \times P(9) \\ &= 0.00 + 0.06 + 0.22 + 0.24 + 0.28 + 0.25 + 0.18 + 0.21 + 0.16 + 0.09 = \\ &= 1.69 \end{aligned}$$

Standard deviation, σ

Greek letters refer to parameters in the population.
Contrast to s (Ch 2, p96)

- Indicates the amount of variability or dispersion in the population
- If you randomly sample from the population, σ describes how far, on average, the sampled value will be from the mean of the distribution
- No formula yet, you'll get a formula for a specific distribution later this lecture

x	P(x)
0	0.54
1	0.06
2	0.11
3	0.08
4	0.07
5	0.05
6	0.03
7	0.03
8	0.02
9	0.01

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Continuous variables

- Continuous variables that can take any decimal value
 - e.g., time to solve an arithmetic item (10.71 s, 13.83 s, etc)
 - e.g., height (171.19 cm, 156.53 cm, etc)
 - e.g., weight babies (3462.83 gr, 4362.17 gr, etc)



Picture source: pxabay.com

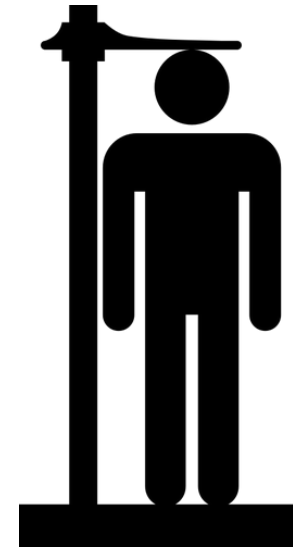
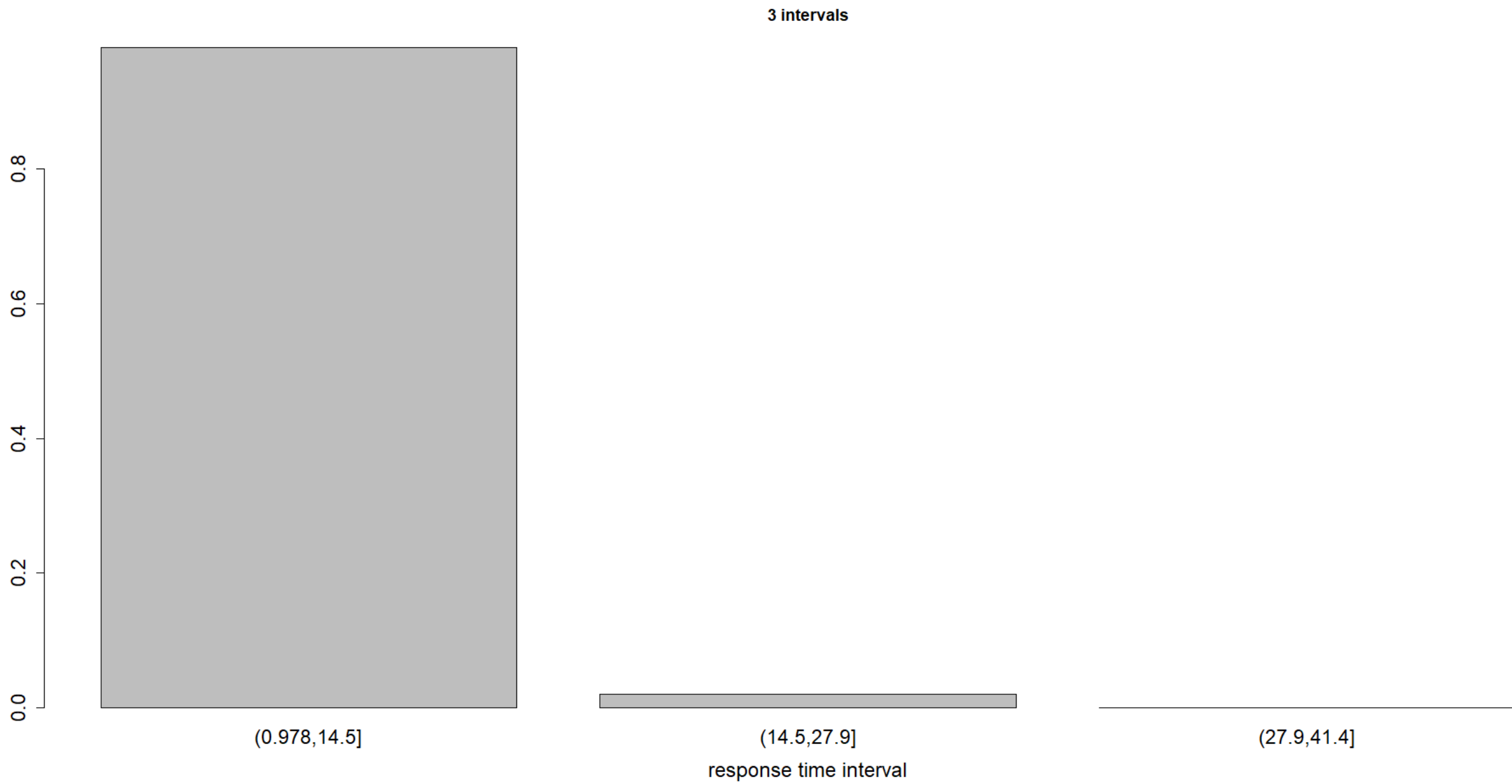
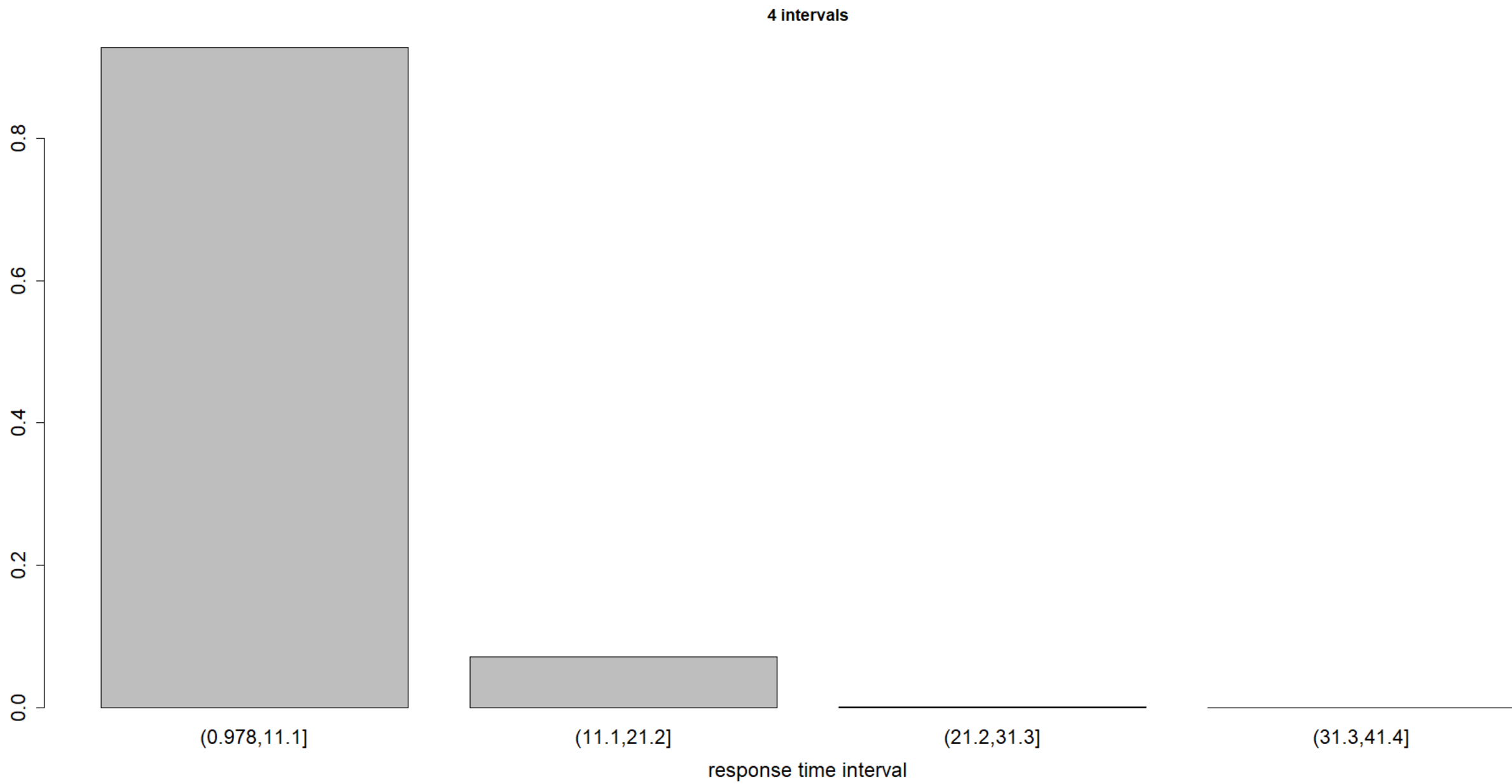


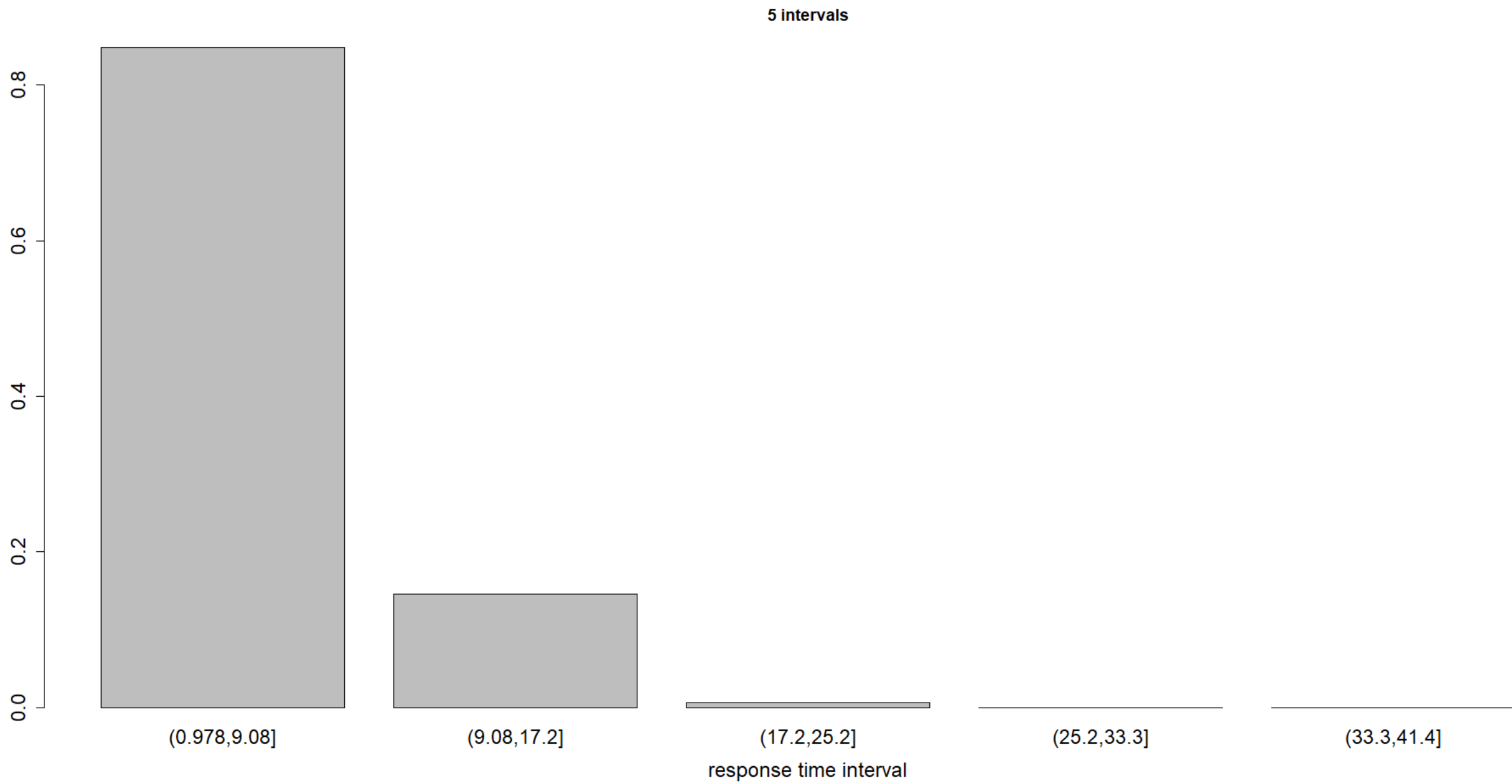
Photo source: <https://publicdomainvectors.org/en/free-clipart/Height-measurement/64875.html>

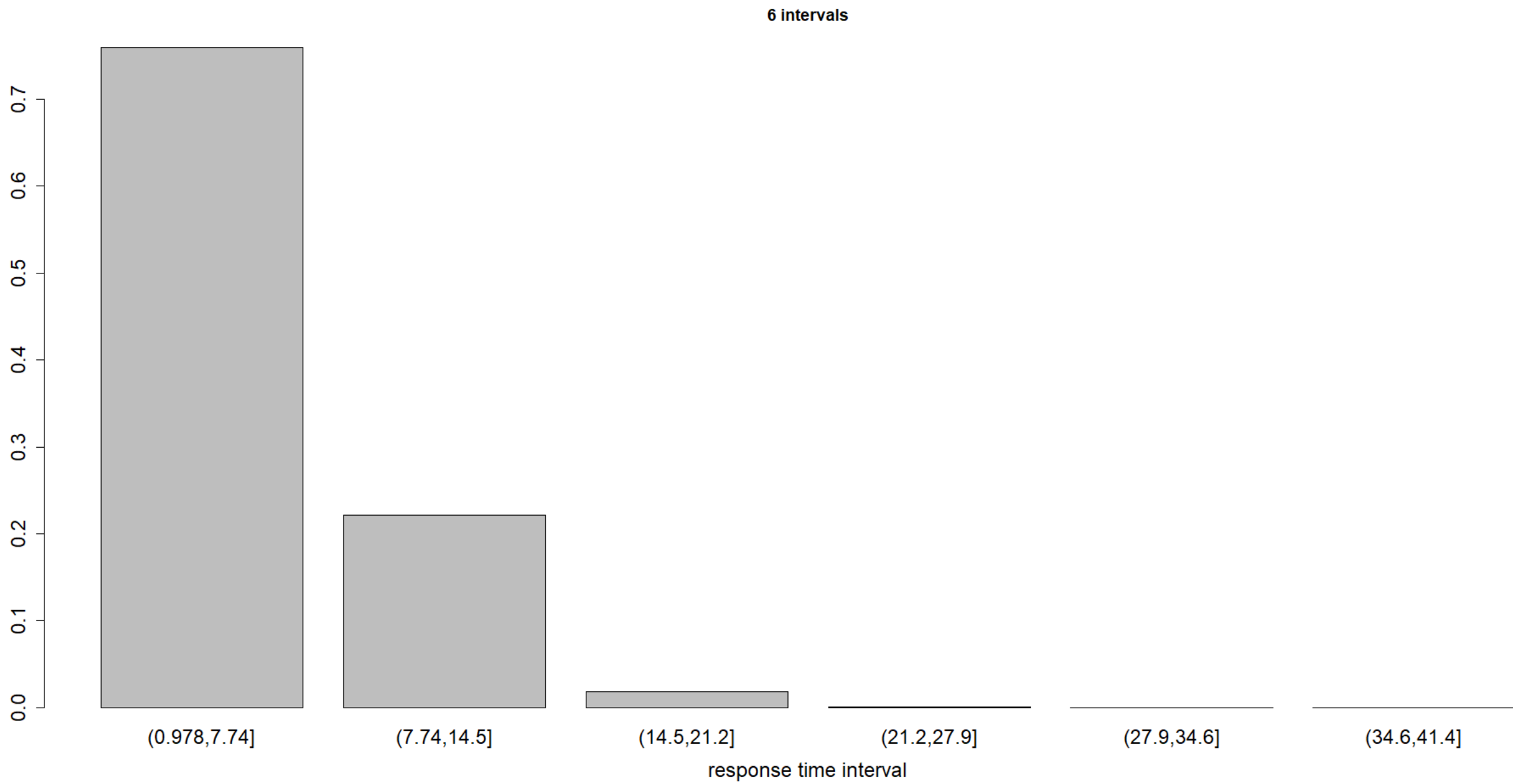
Example continuous variable: Time to solve an arithmetic item

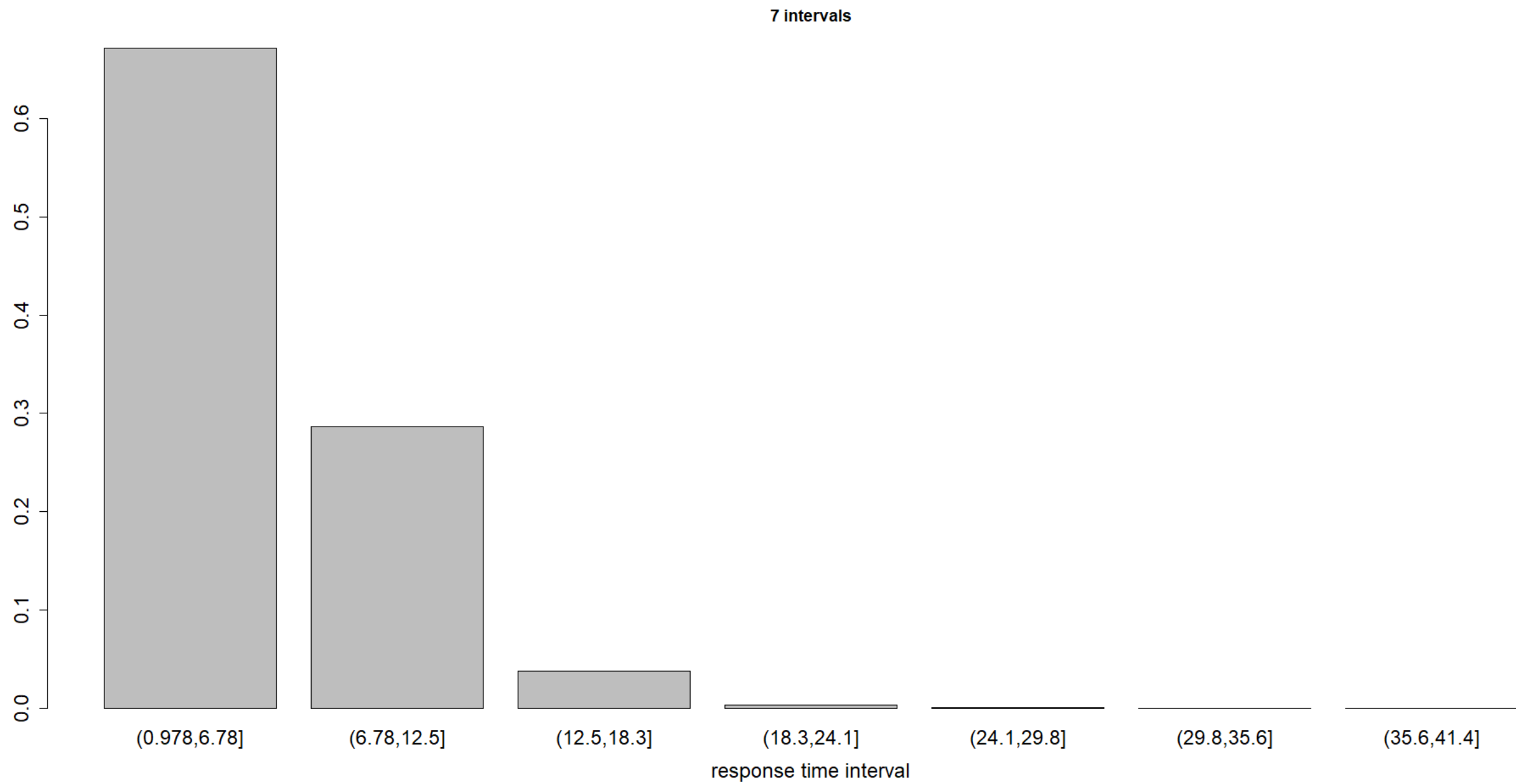
Subject	Response time (s)
1	13.46649
2	2.662474
3	8.177177
4	1.832033
5	5.550982
6	4.835395
7	3.124351
...	
1000	5.9045371

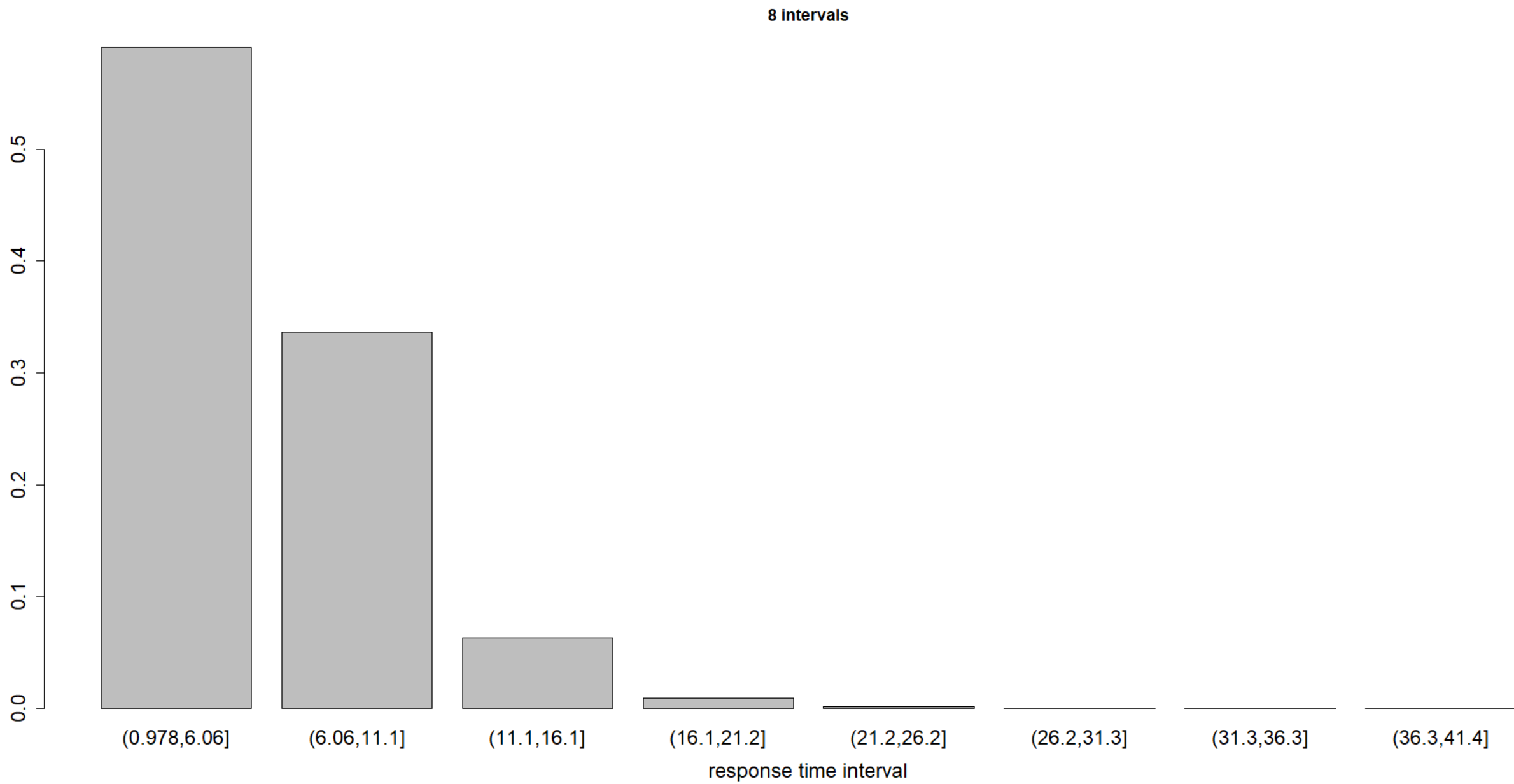


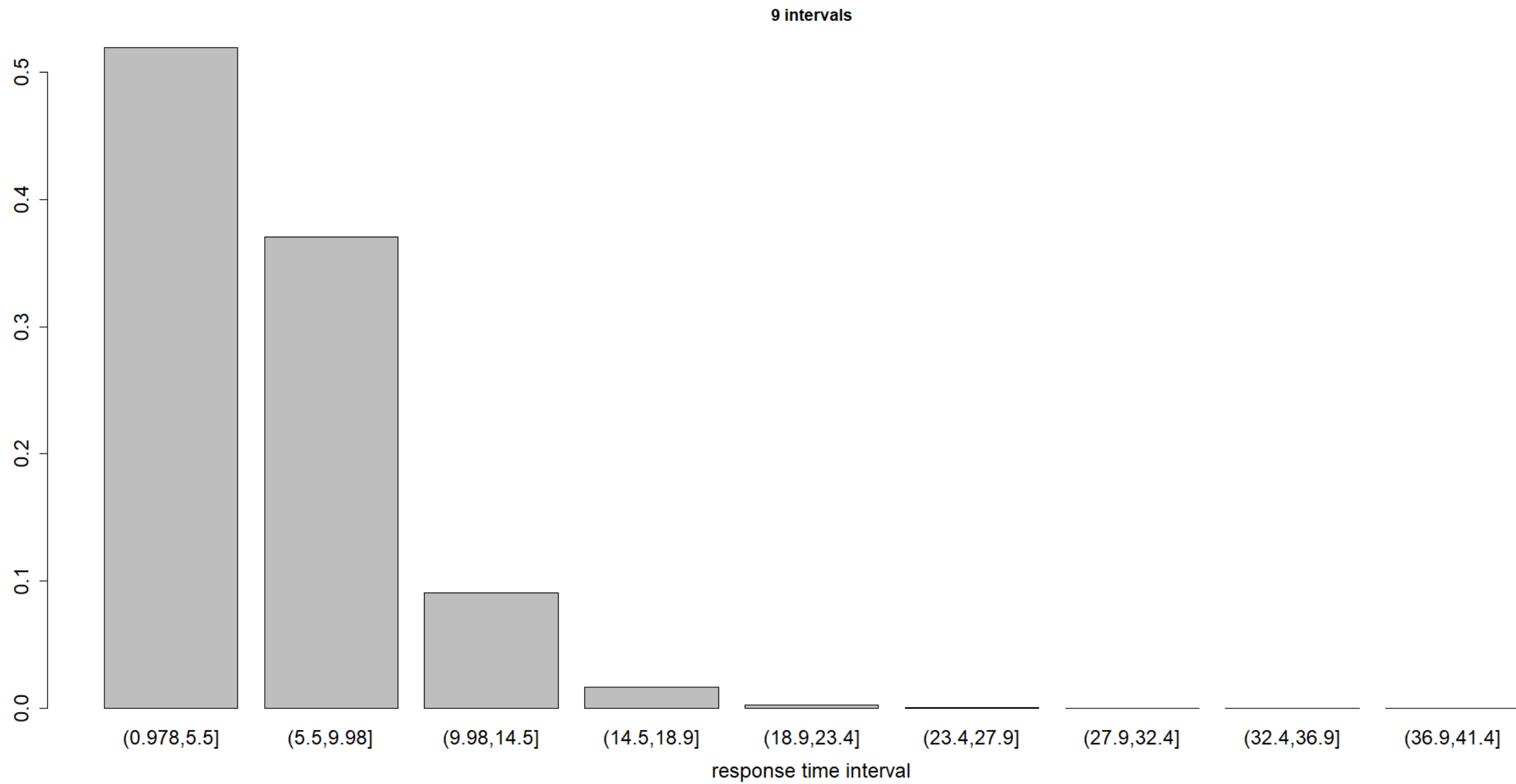


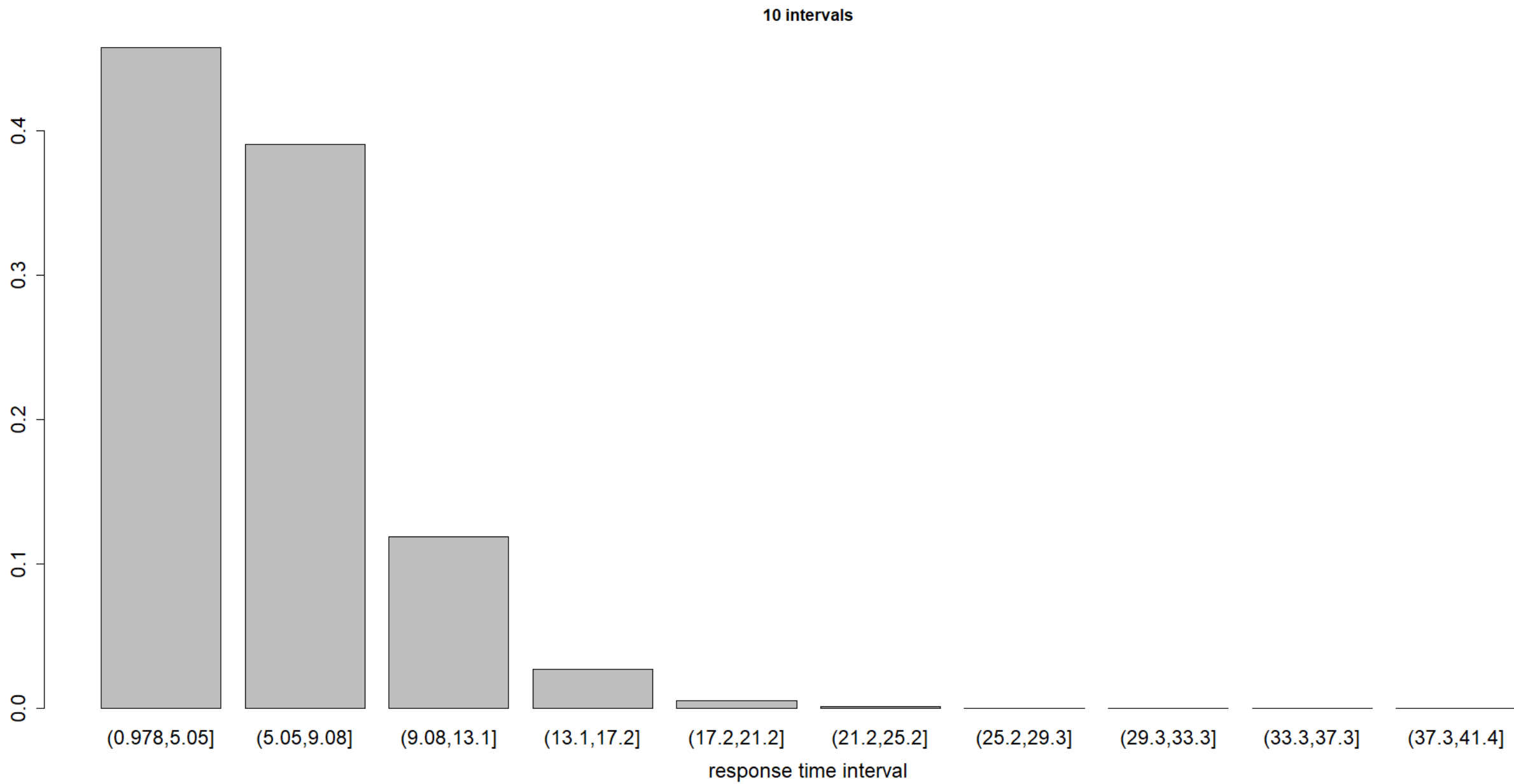




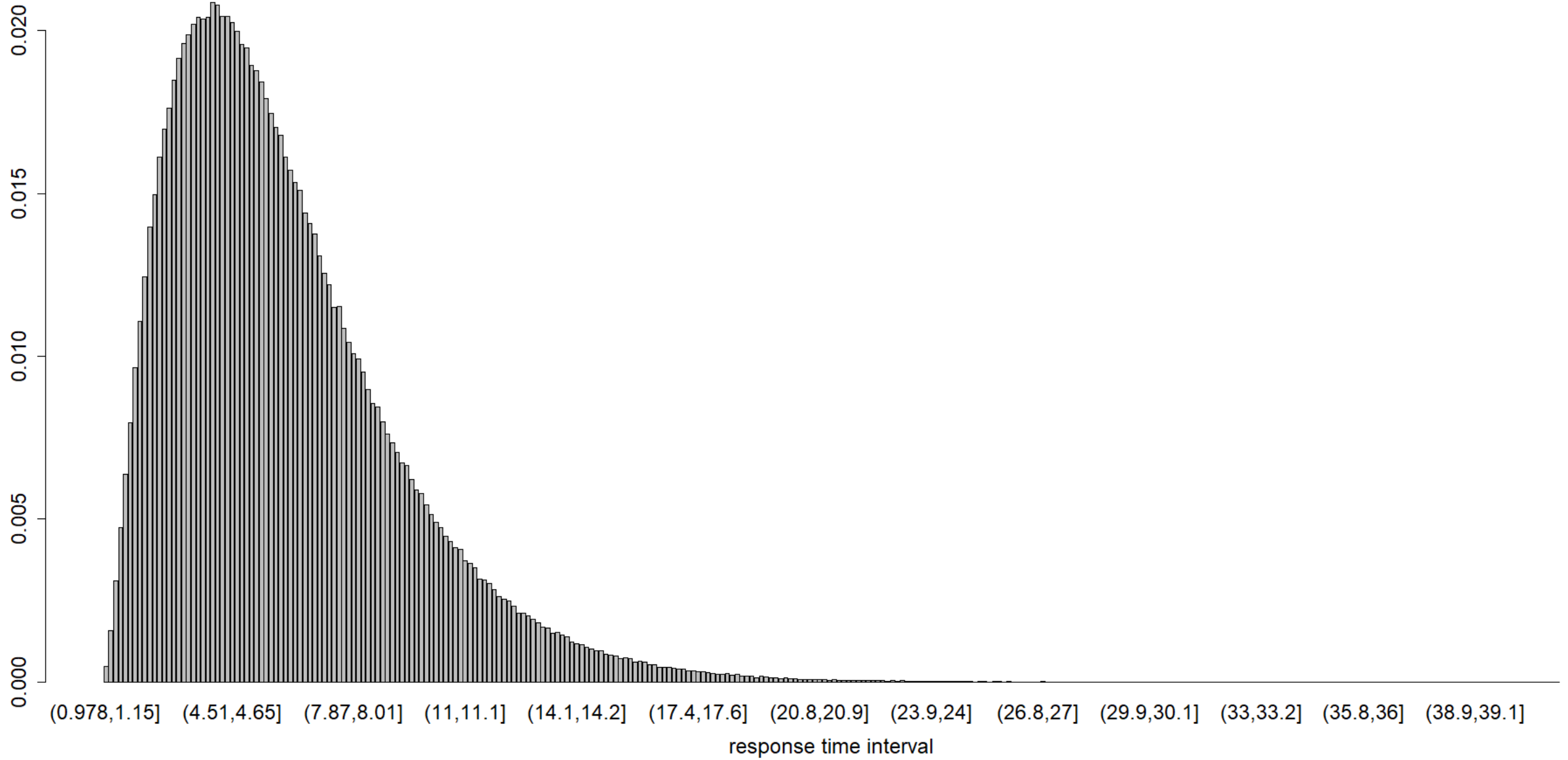


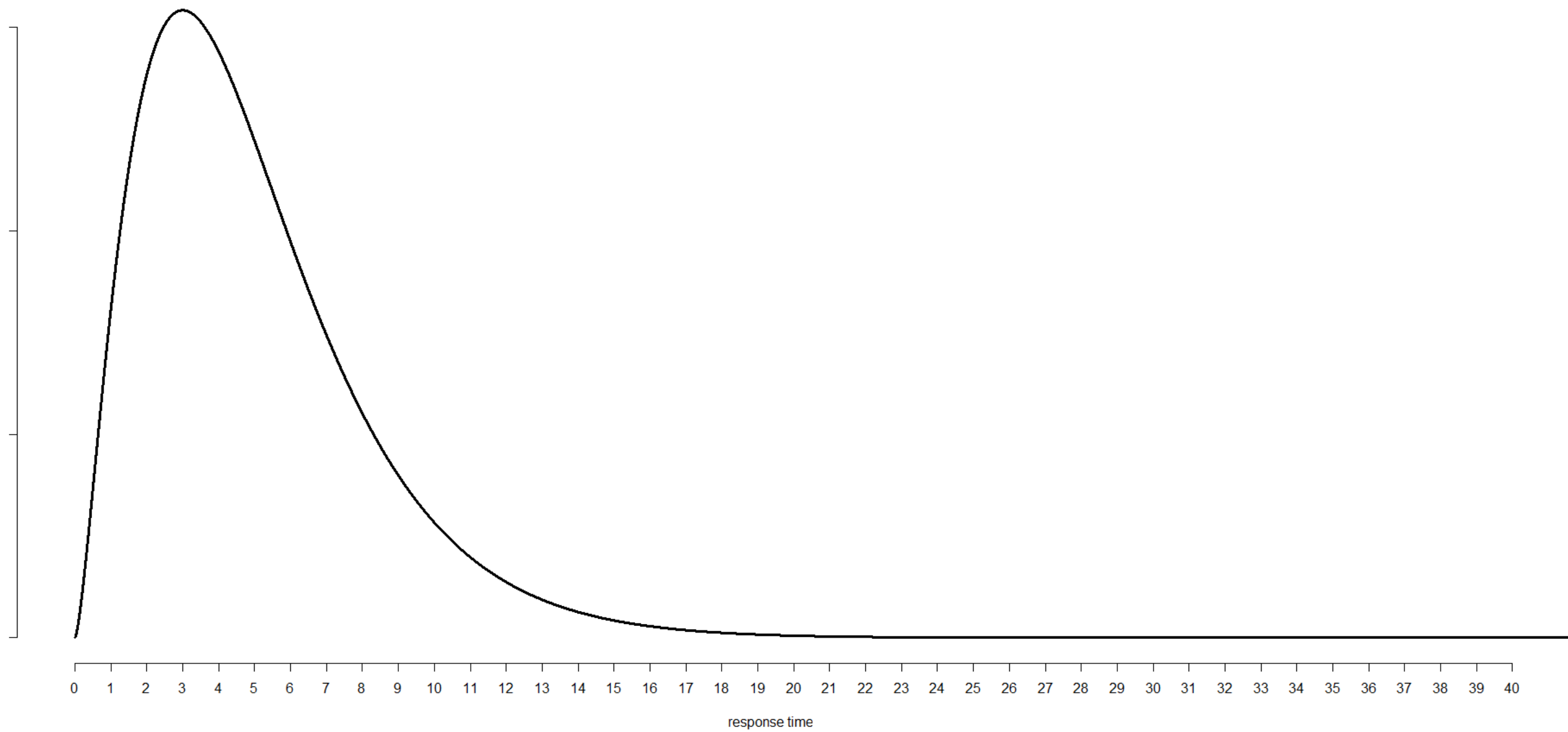


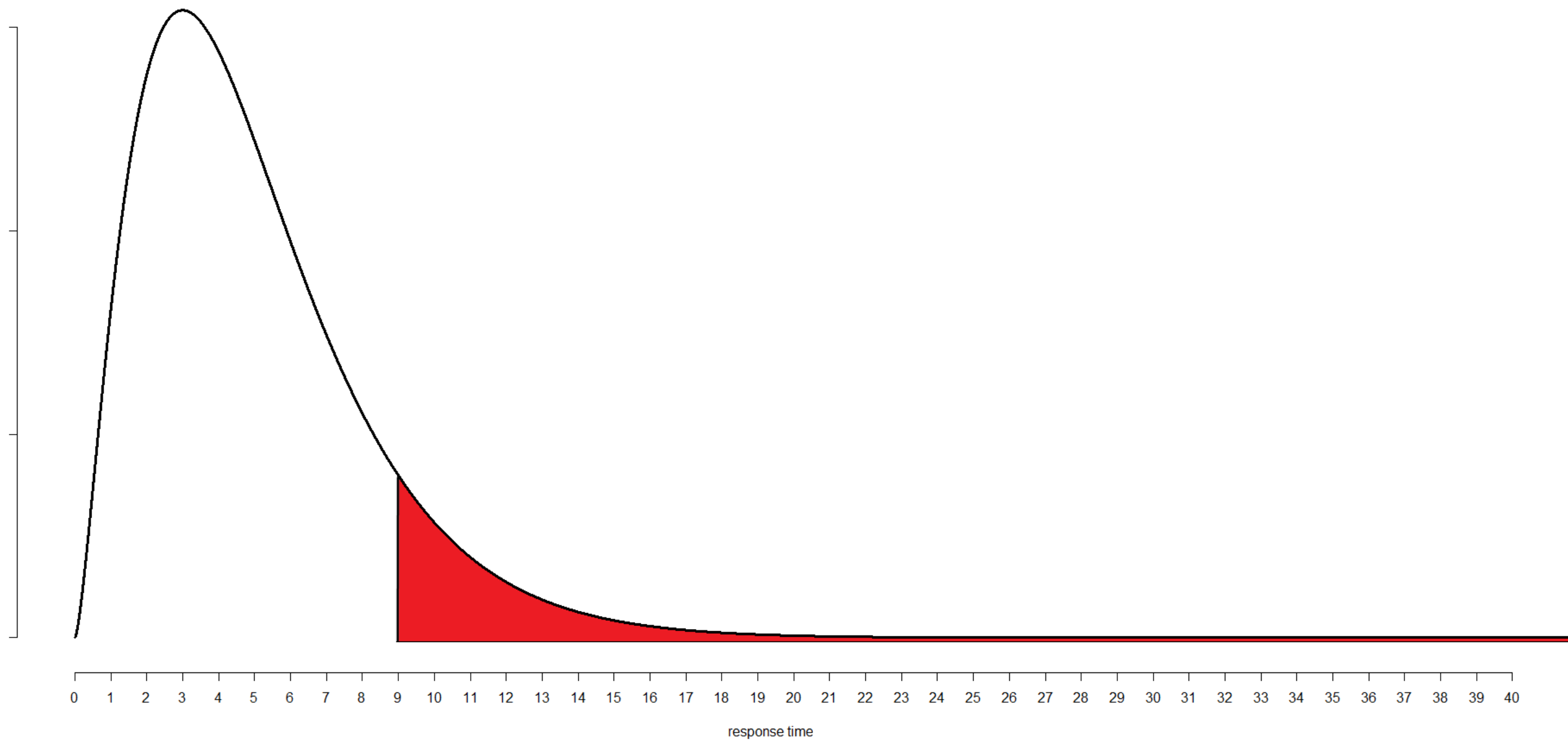


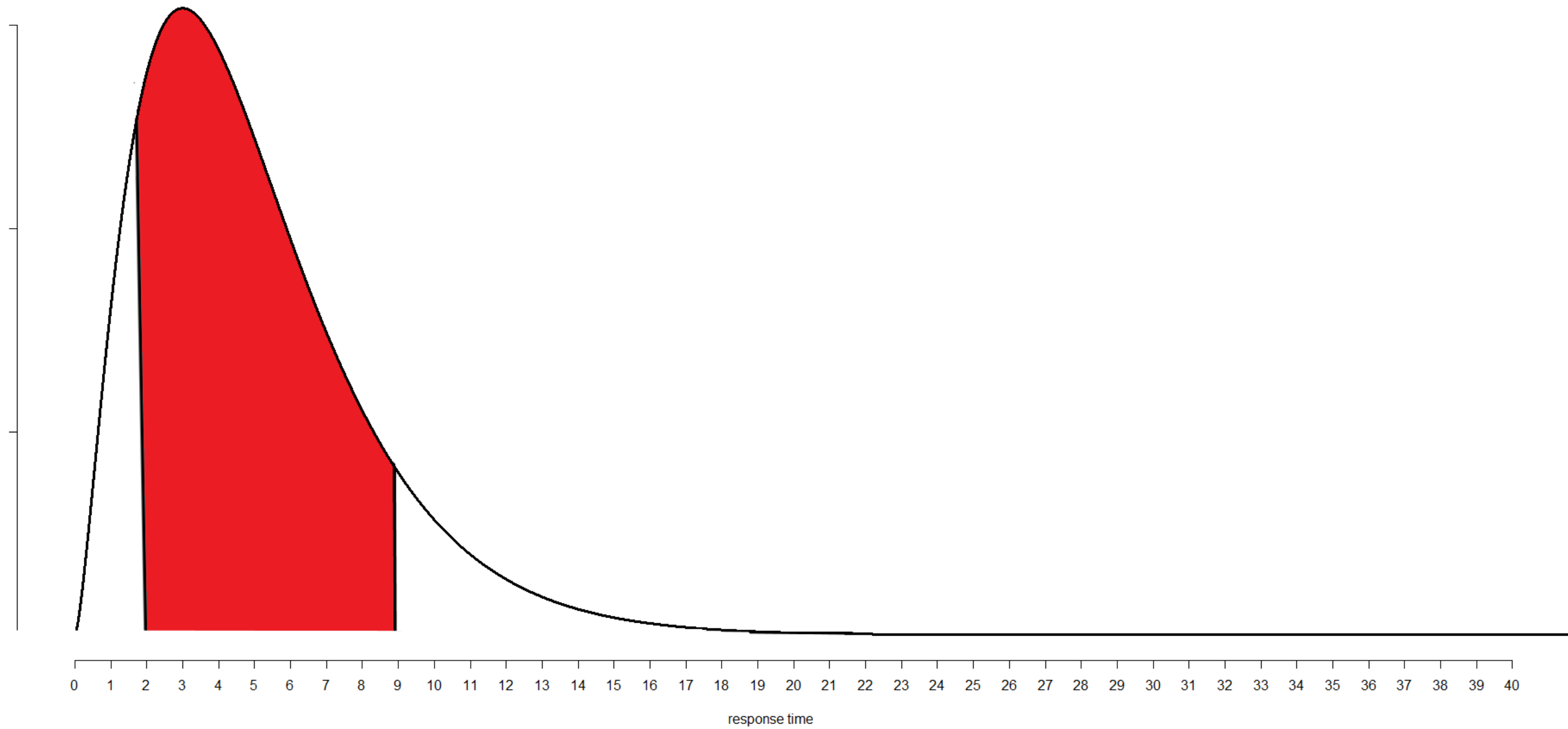


300 intervals



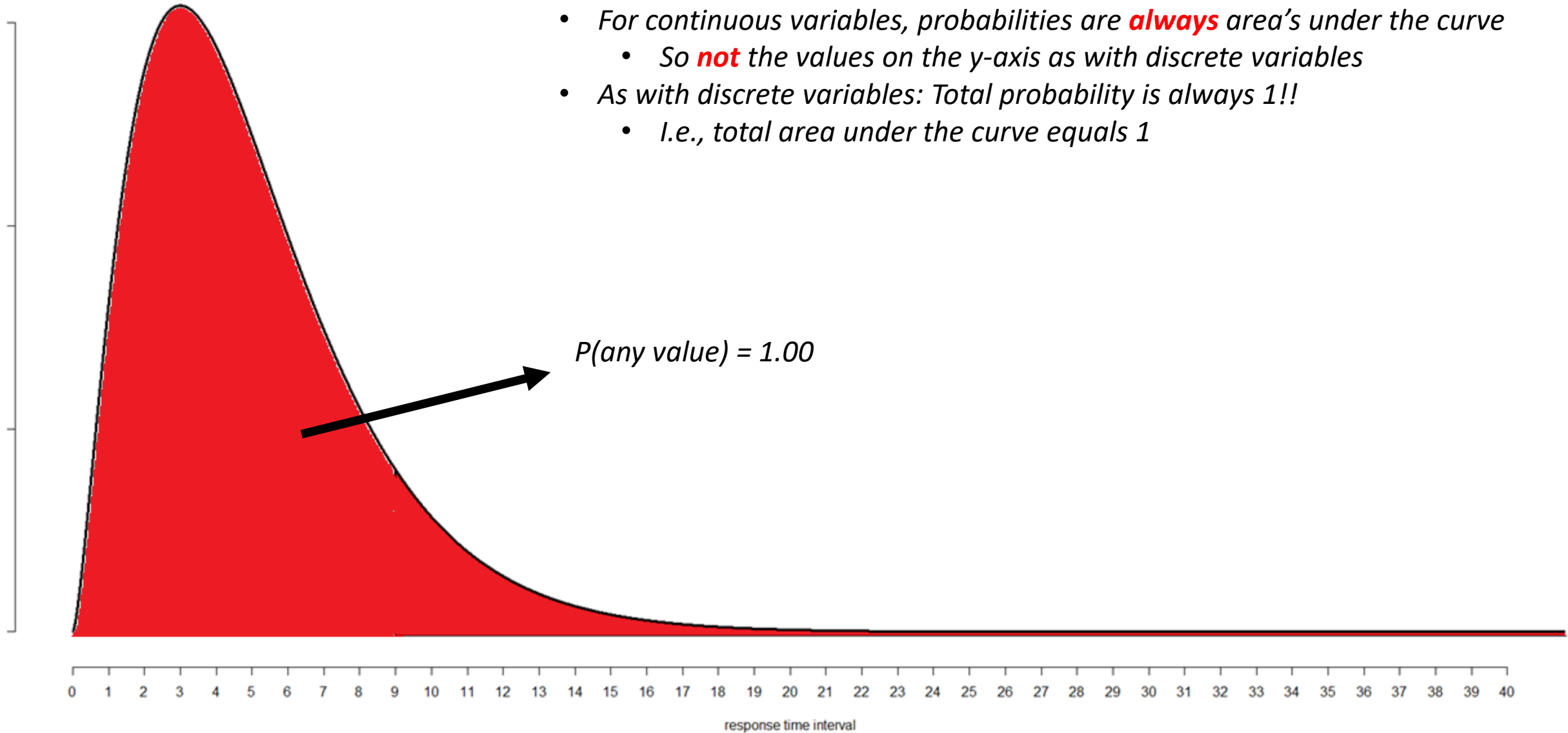






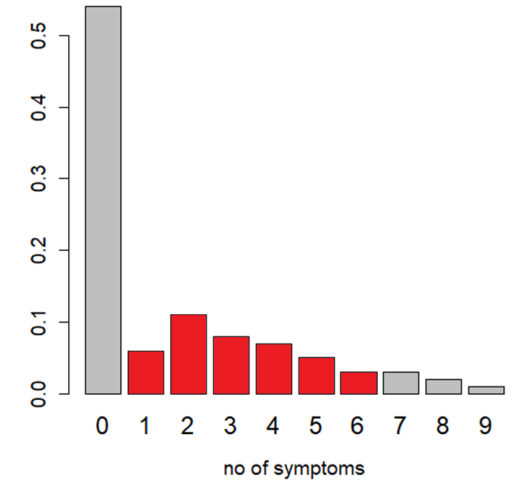
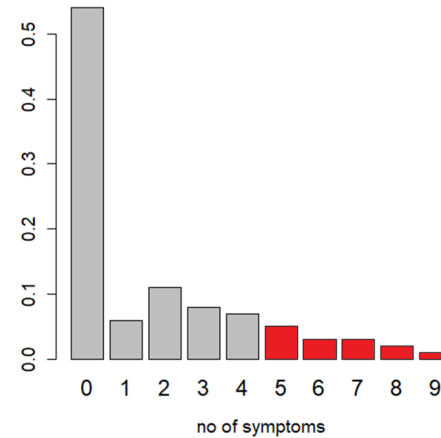
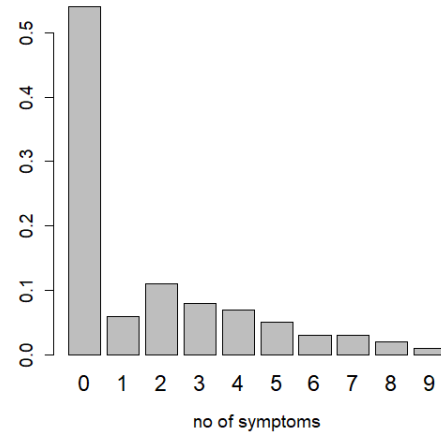
Some notes:

- For continuous variables, probabilities are **always** area's under the curve
 - So **not** the values on the y-axis as with discrete variables
- As with discrete variables: Total probability is always 1!!
 - I.e., total area under the curve equals 1

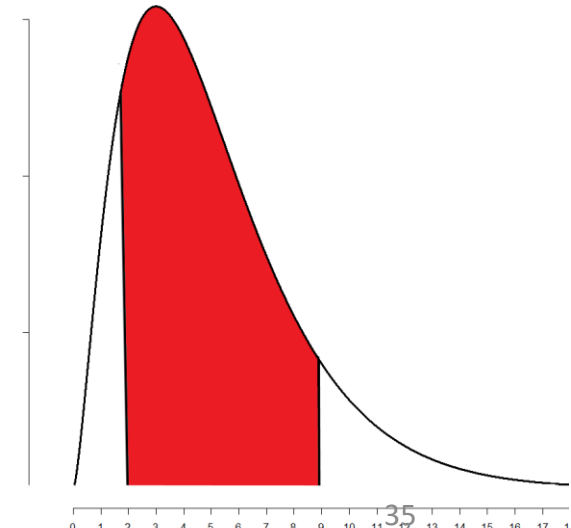
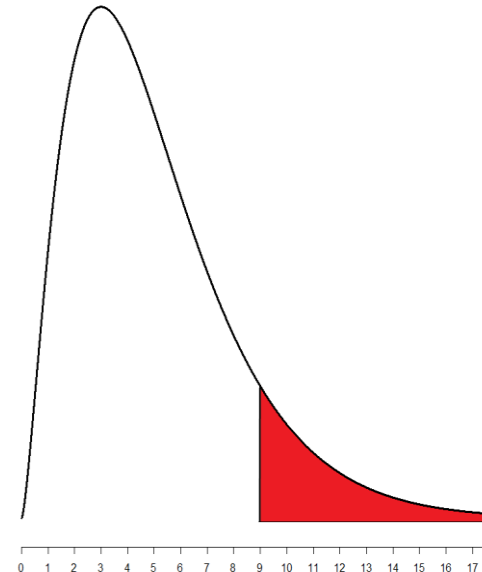
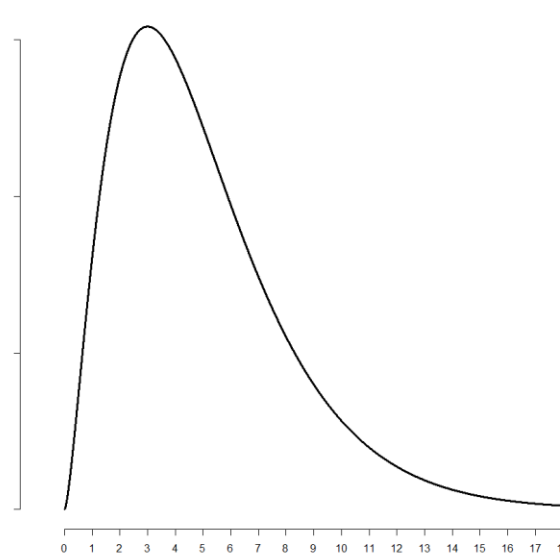


To summarize:

Discrete Distributions



Continuous Distributions



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The normal distribution

“Bell-shaped curve”

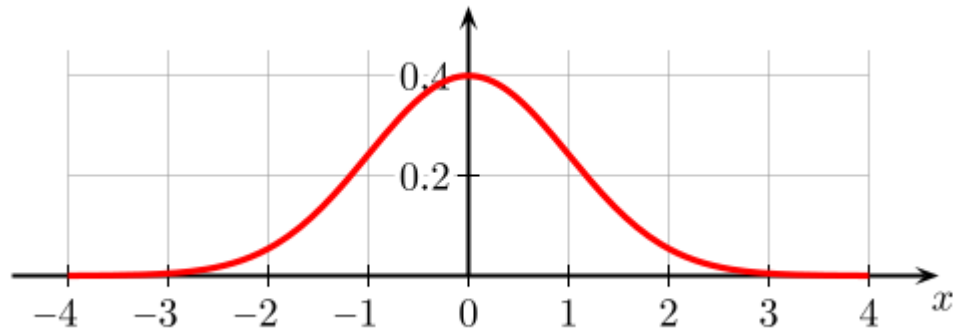
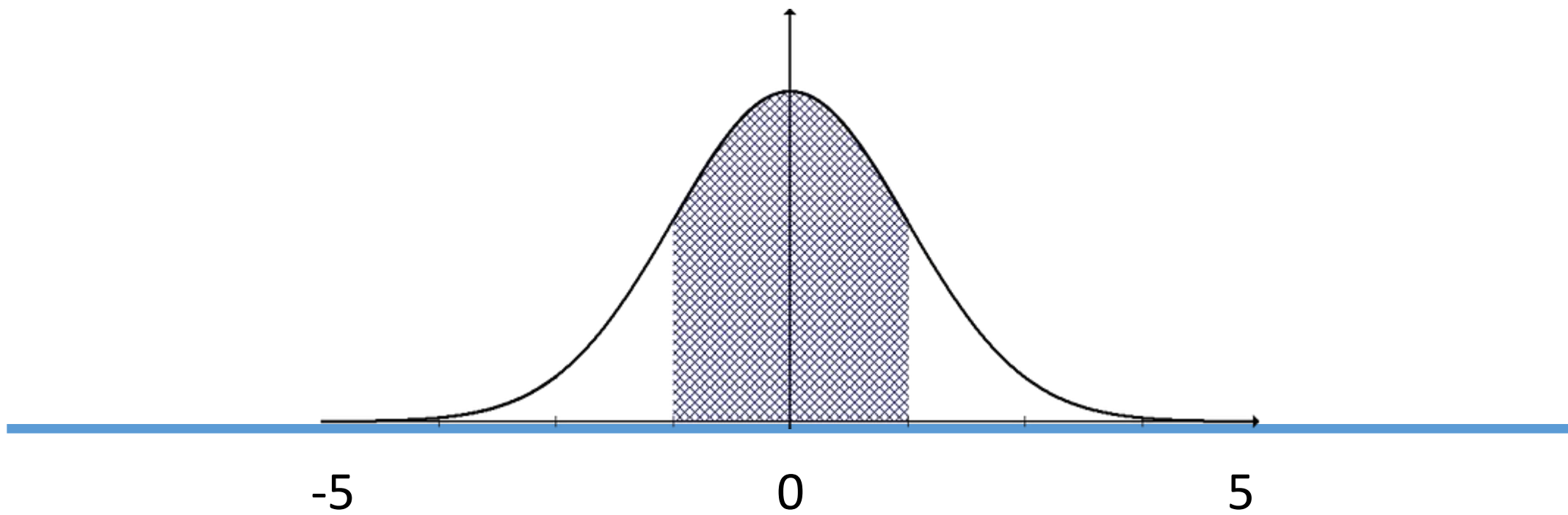
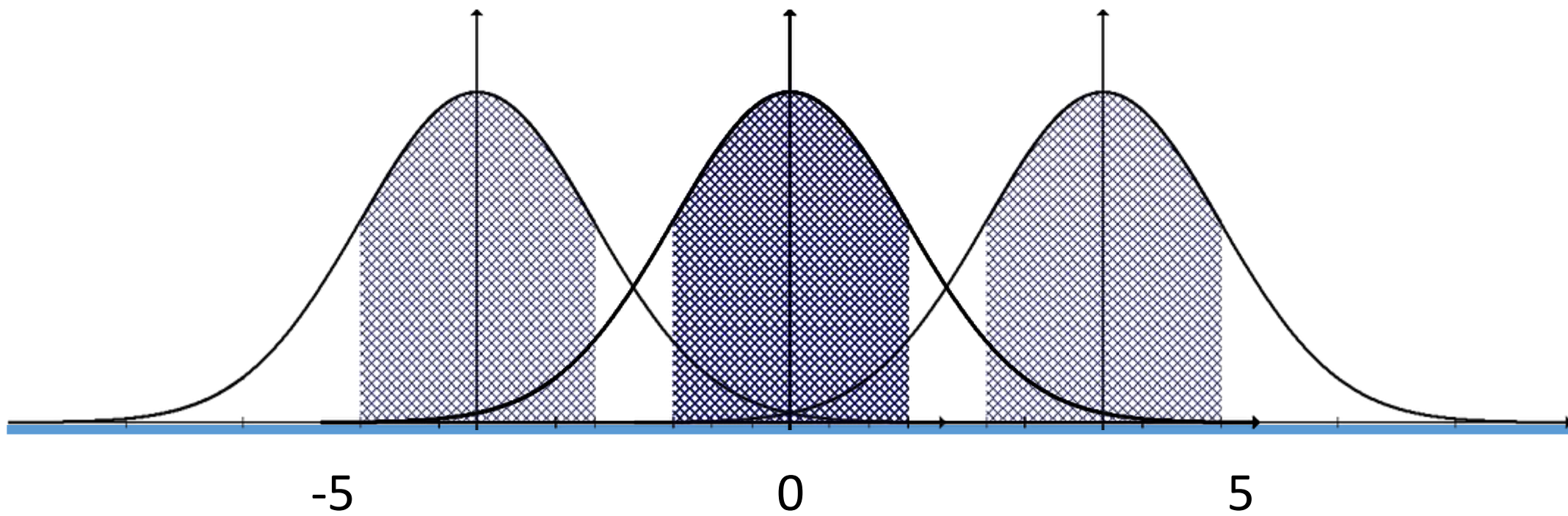


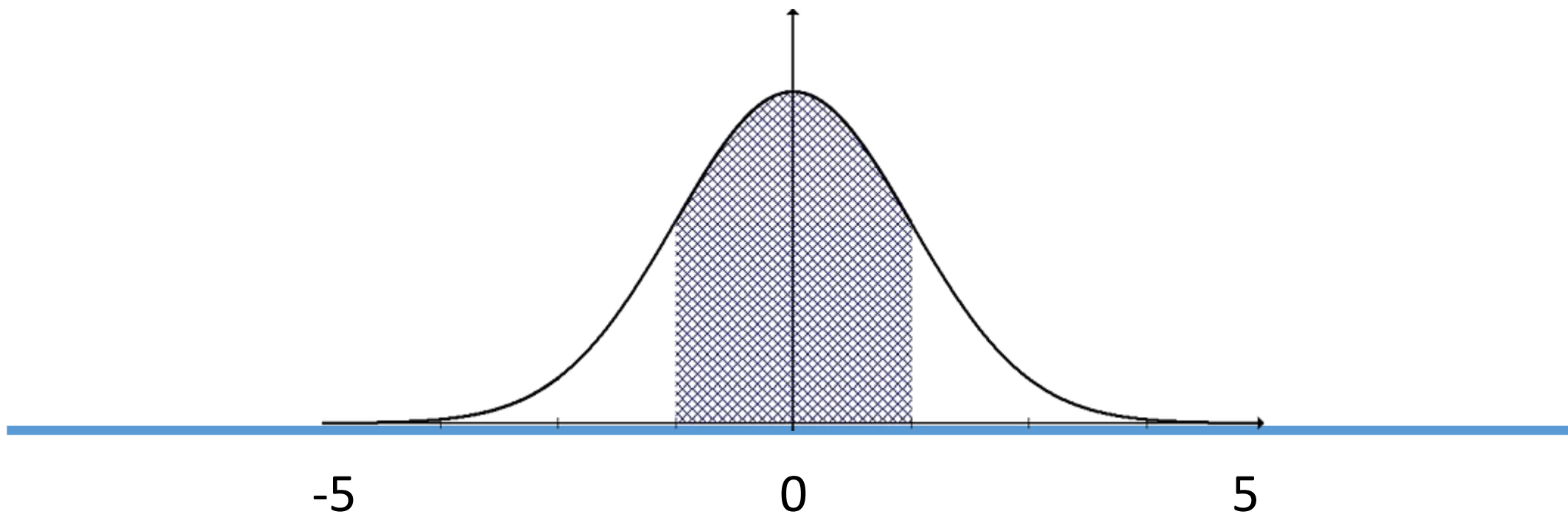
Photo source: https://commons.wikimedia.org/wiki/File:Normal_distribution.svg

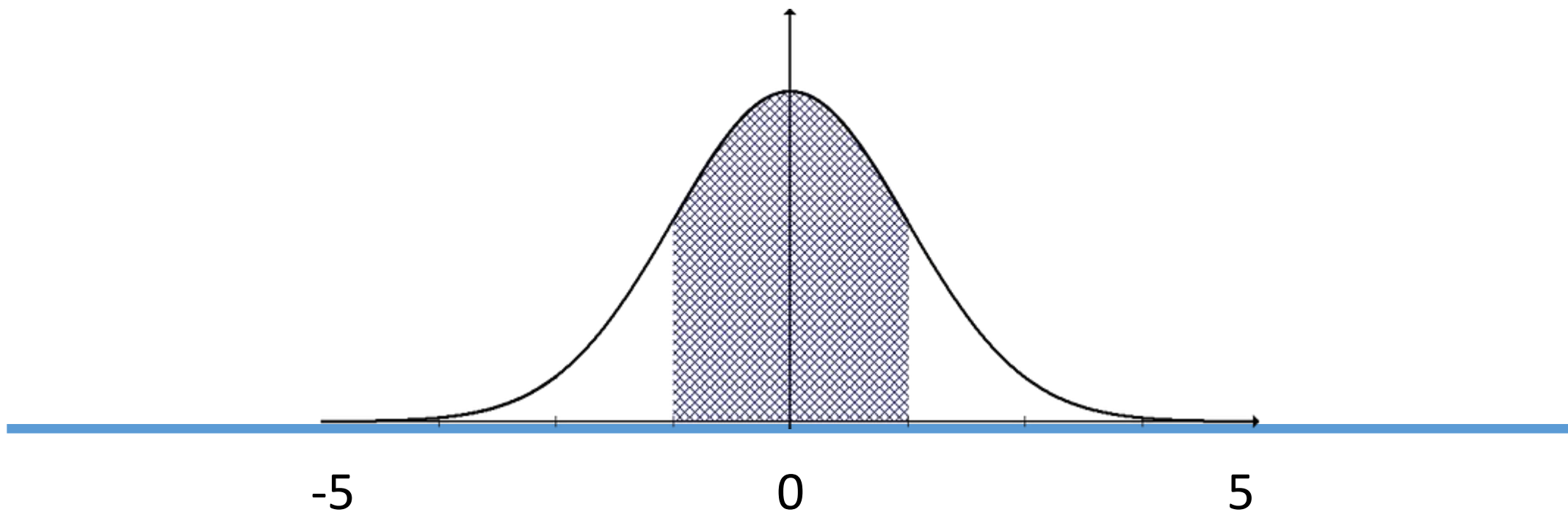


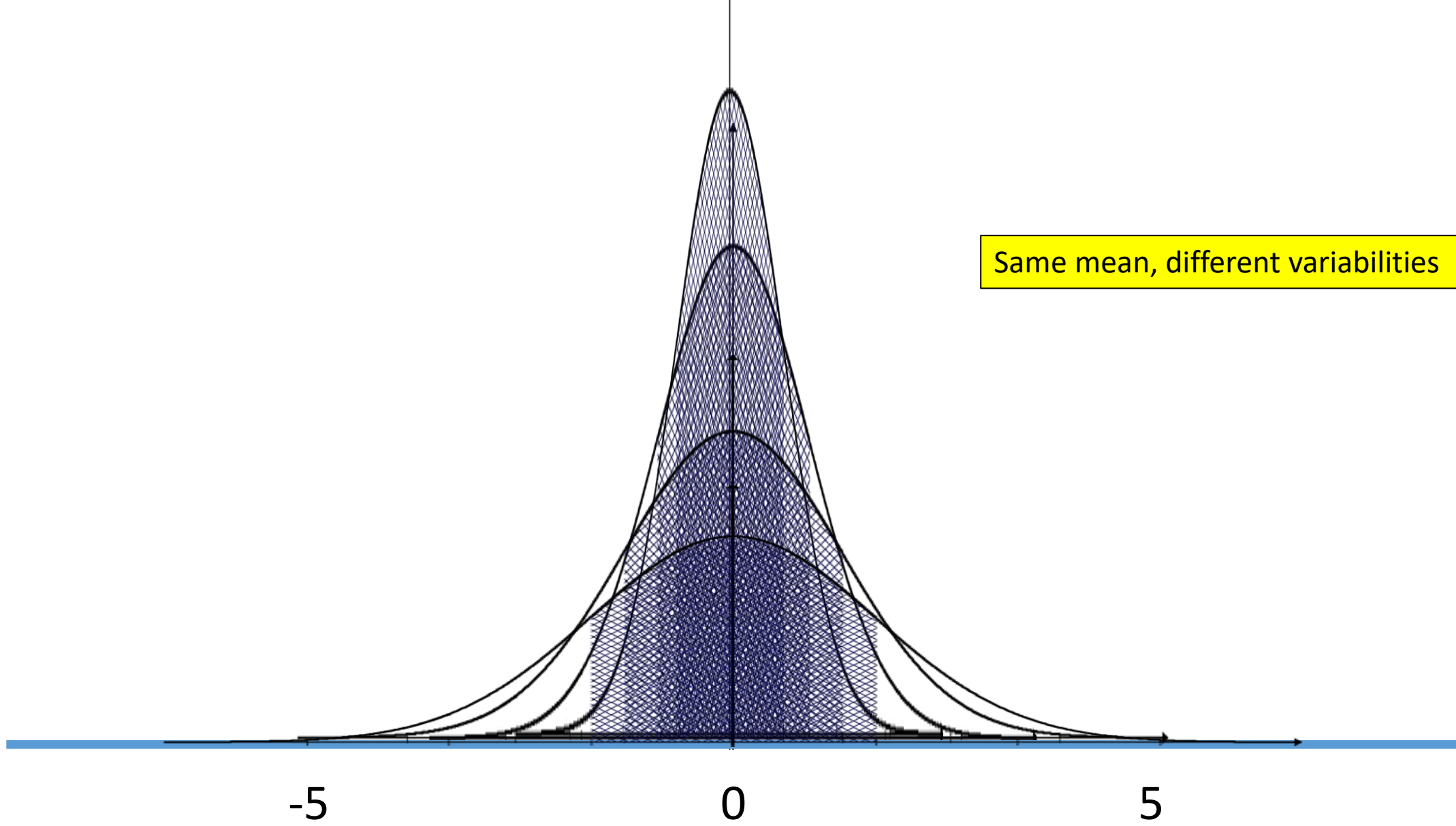
Photo source: https://commons.wikimedia.org/wiki/File:Liberty_Bell_icon.svg



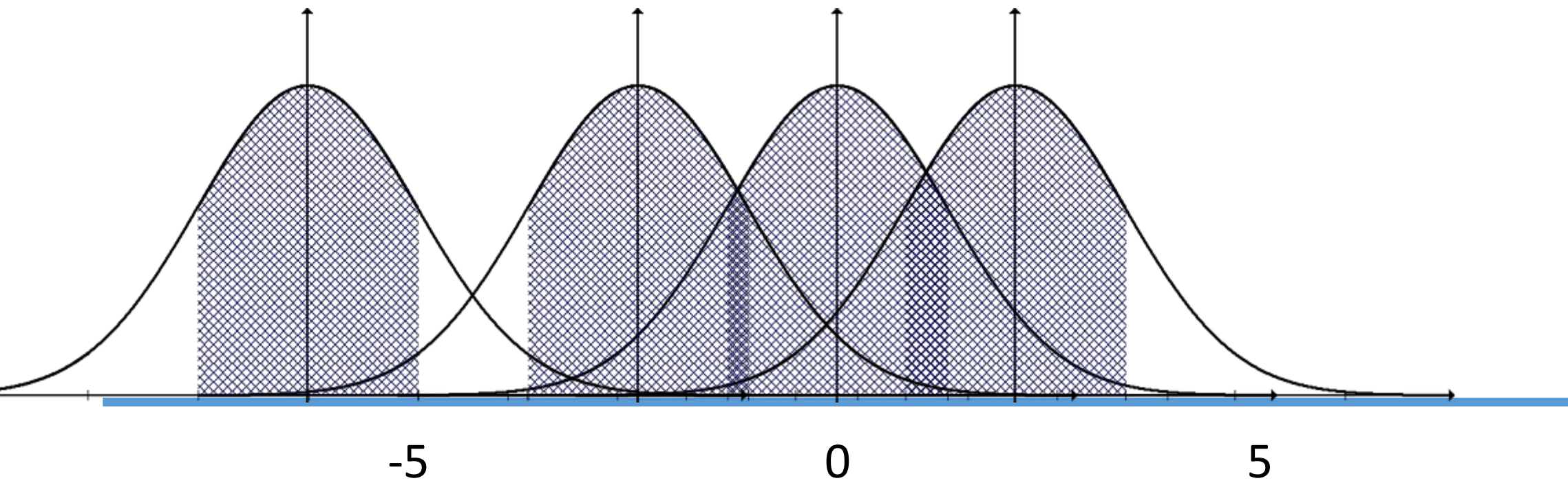


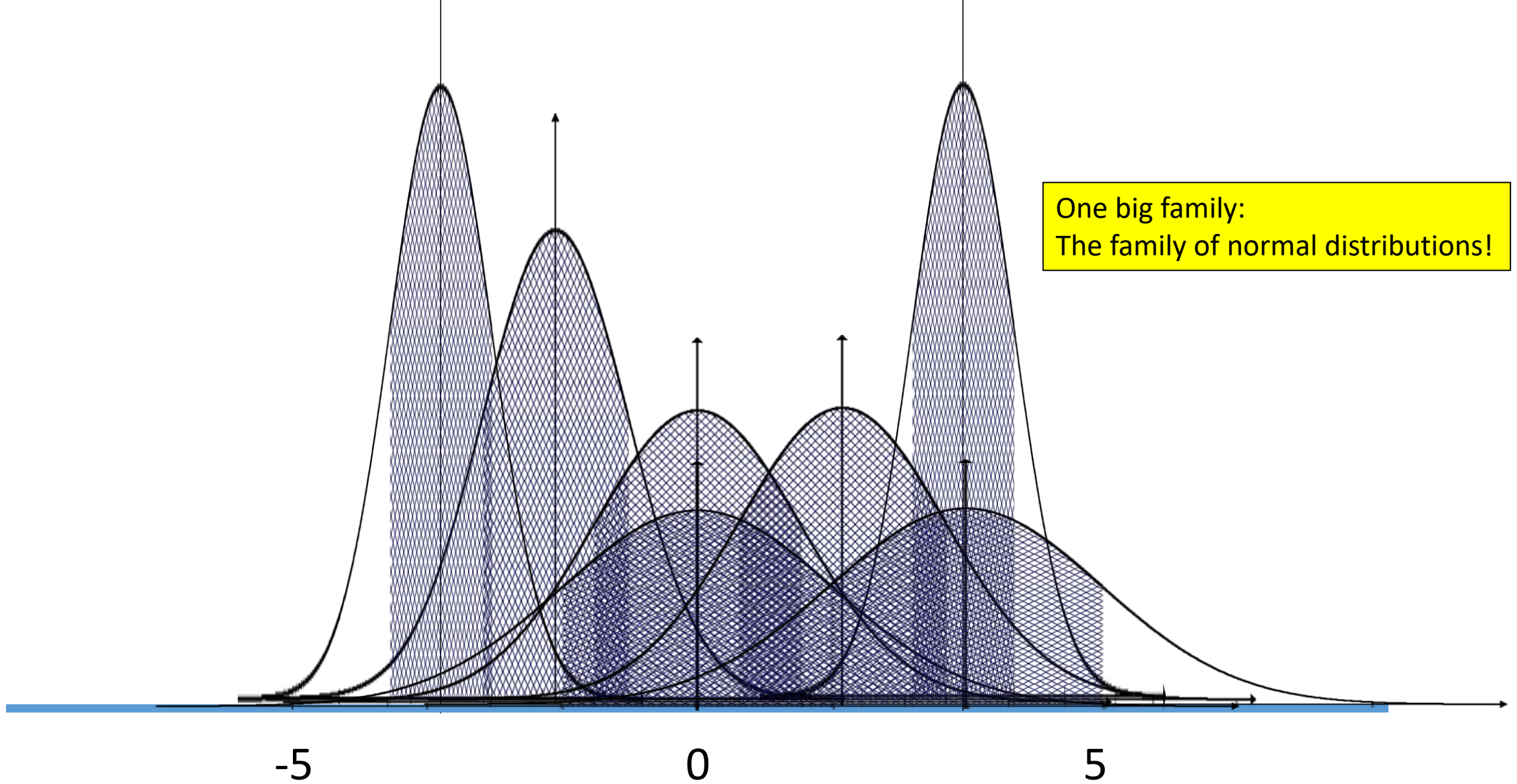






Same variability, different means





The normal distribution is characterized by 2 parameters: the mean and Standard Deviation. This means that if you know the values of these two parameters (e.g., mean = 2 and SD = 1) then you identify one single curve from the family.

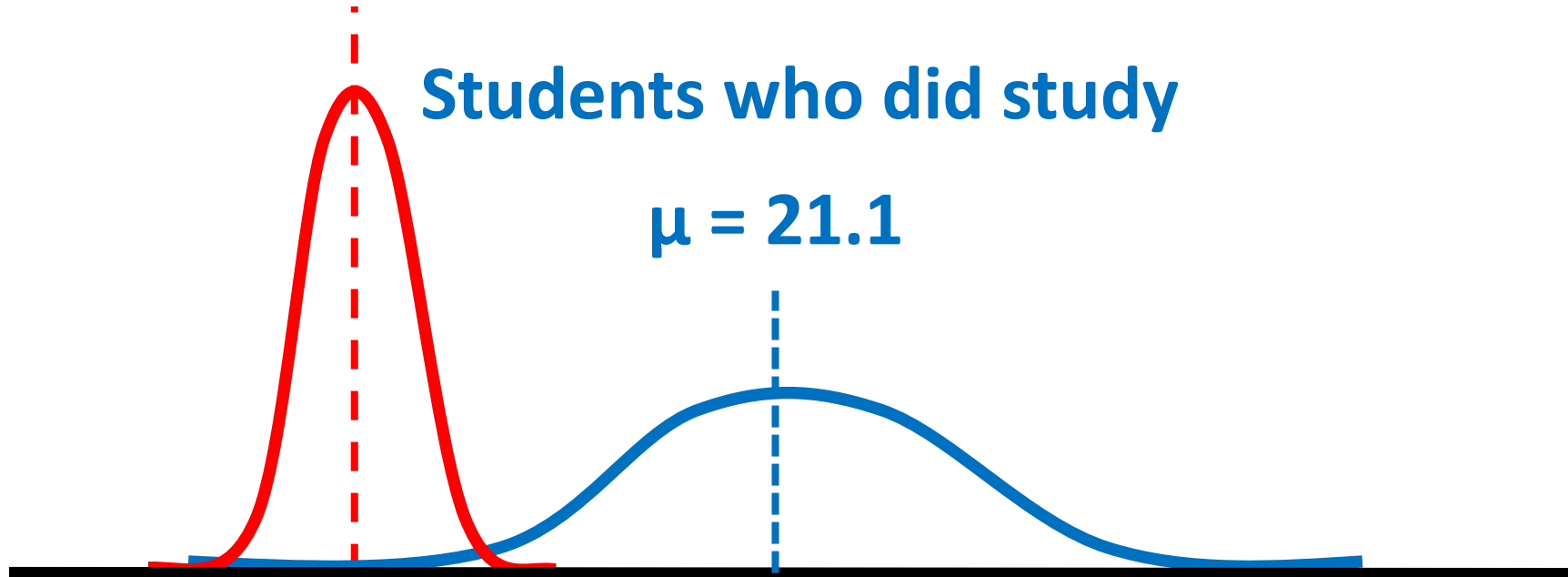
The normal distribution: mean μ

Students who didn't study

$$\mu = 8.3$$

Students who did study

$$\mu = 21.1$$



Score interim exam

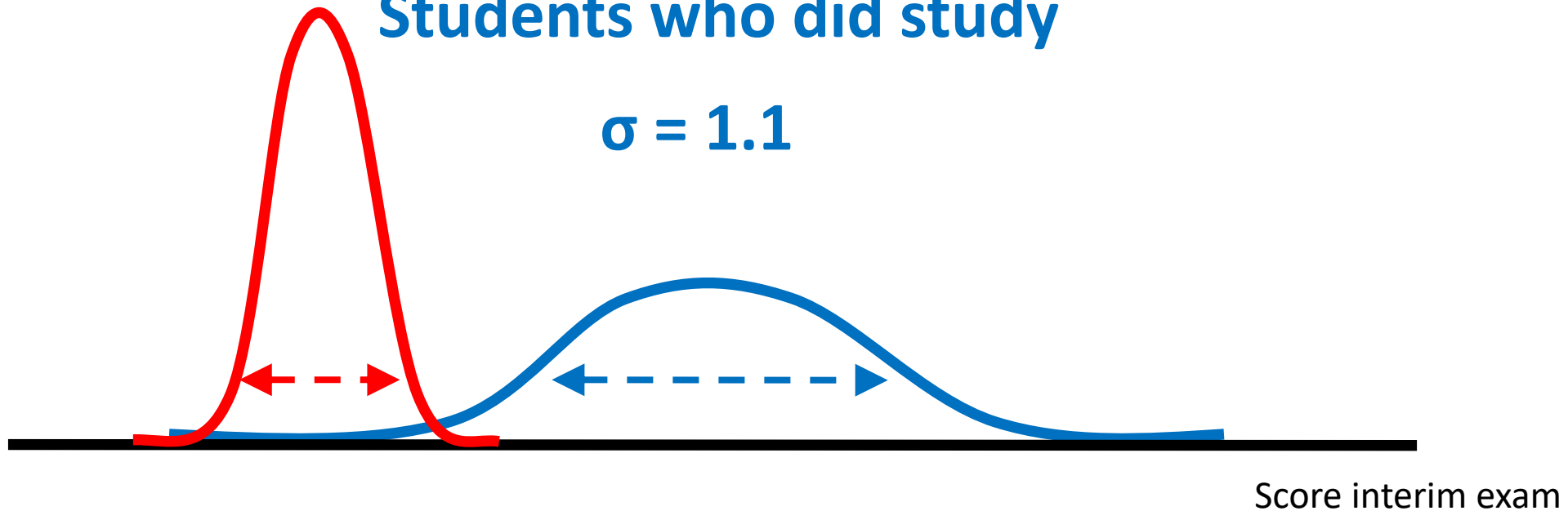
The normal distribution: standard deviations σ

Students who didn't study

$$\sigma = 0.3$$

Students who did study

$$\sigma = 1.1$$



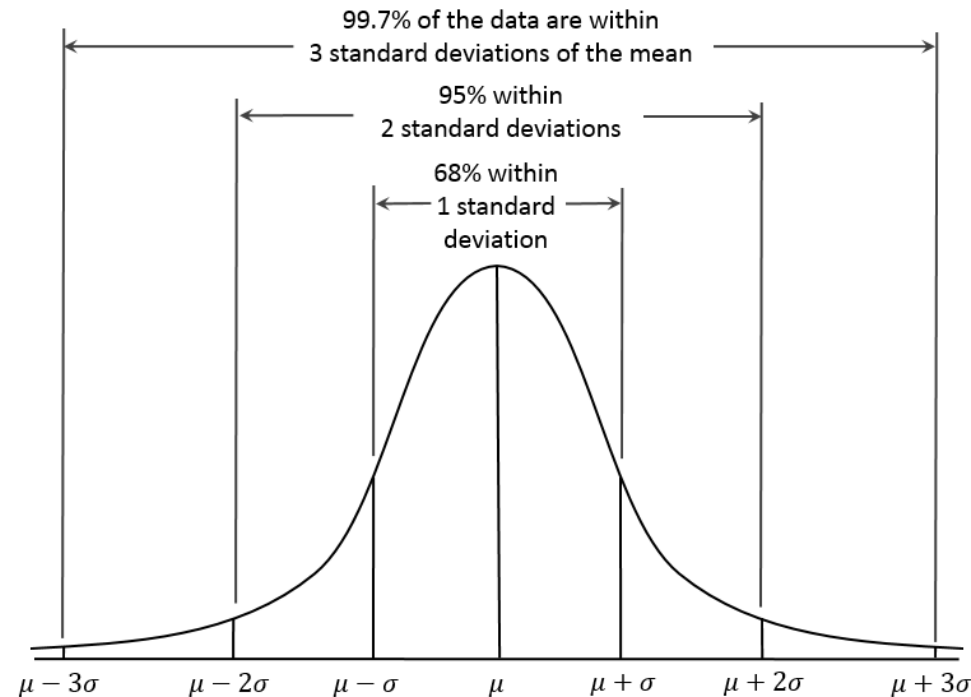
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 - Three different probability calculations
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- Specific distribution for continuous variables: Normal distribution
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 - **Three different probability calculations**
- Specific distribution for discrete variables: Binomial distribution
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Probability calculation

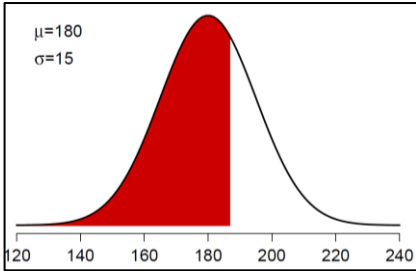
Empirical rule (section 2.4; section 6.2)

- Can be used for rough probability calculation



- But in the probability calculations you will practice with, you will use more precise calculations

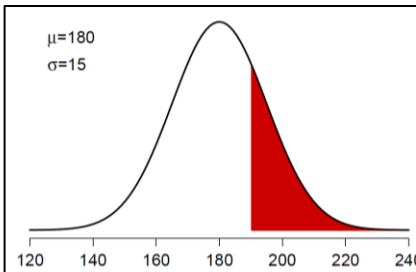
Three different probability calculations



1) Probability of reaction time shorter than 187.5

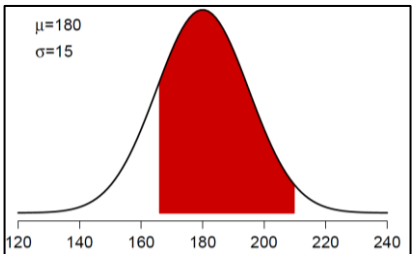
- $P(X < 187.5)$

In continuous distributions, we use $<$ and $>$ (not $\leq$ and $\geq$) because the result is the same e.g., $P(X < 190) = P(X \leq 190)$. This is not the case for discrete variables!



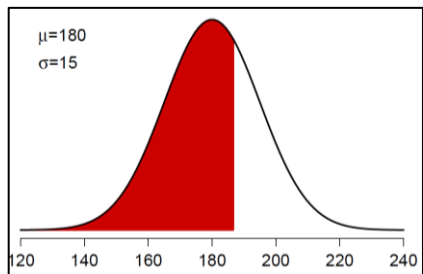
2) Probability of reaction time longer than 190

- $P(X > 190)$



3) Probability of reaction time between 165 and 210

- $P(165 < X < 210)$



Situation I: Lower tail probability

Step 1: Find z

- In Chapter 2 (and lecture 4 by Johnny)

$$z = \frac{\text{observation} - \text{mean}}{\text{standard deviation}}$$

→ z = the number of standard deviations that a given observation falls from the mean

- If we apply this to the normal distribution, we get:

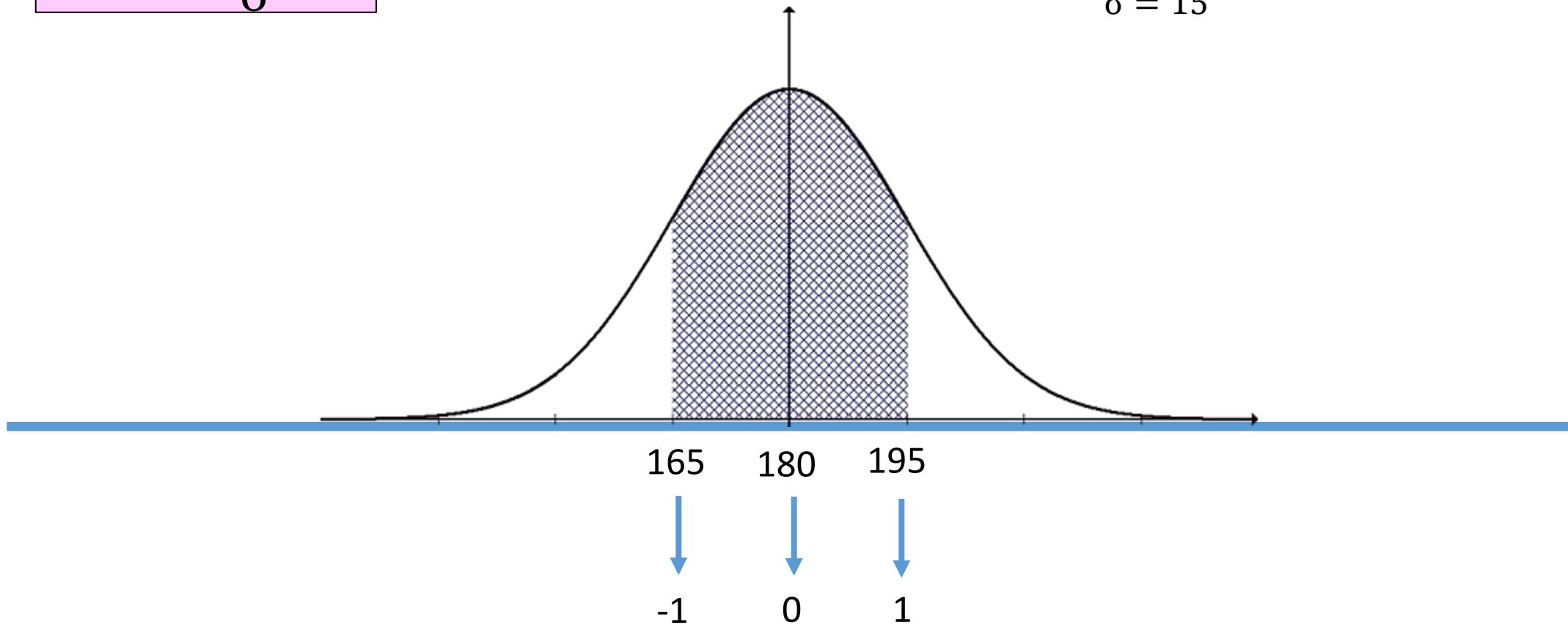
$$z = \frac{x - \mu}{\sigma}$$

where x is an observation (e.g., someone's height, someone's reaction time, etc)

Step 1: Find z

$$z = \frac{x - \mu}{\sigma}$$

$$\mu = 180$$
$$\sigma = 15$$

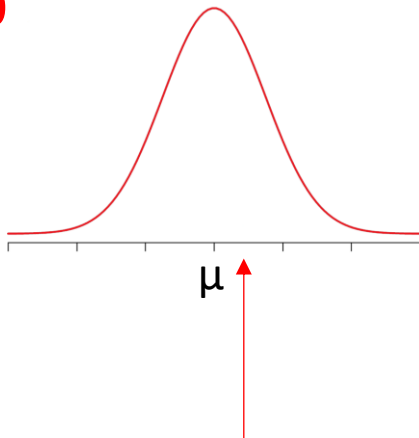


Step 1: Find z

The question was: what is $P(X < 187.5)$?

Original variable (e.g., reaction time)

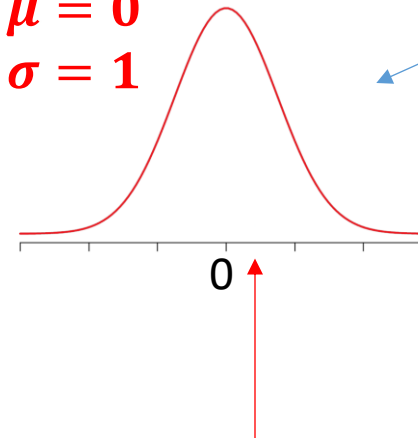
$$\mu = 180$$
$$\sigma = 15$$



$$x = 187.5$$

Corresponding z-score

$$\mu = 0$$
$$\sigma = 1$$



$$z = \frac{187.5 - 180}{15} = 0.5$$

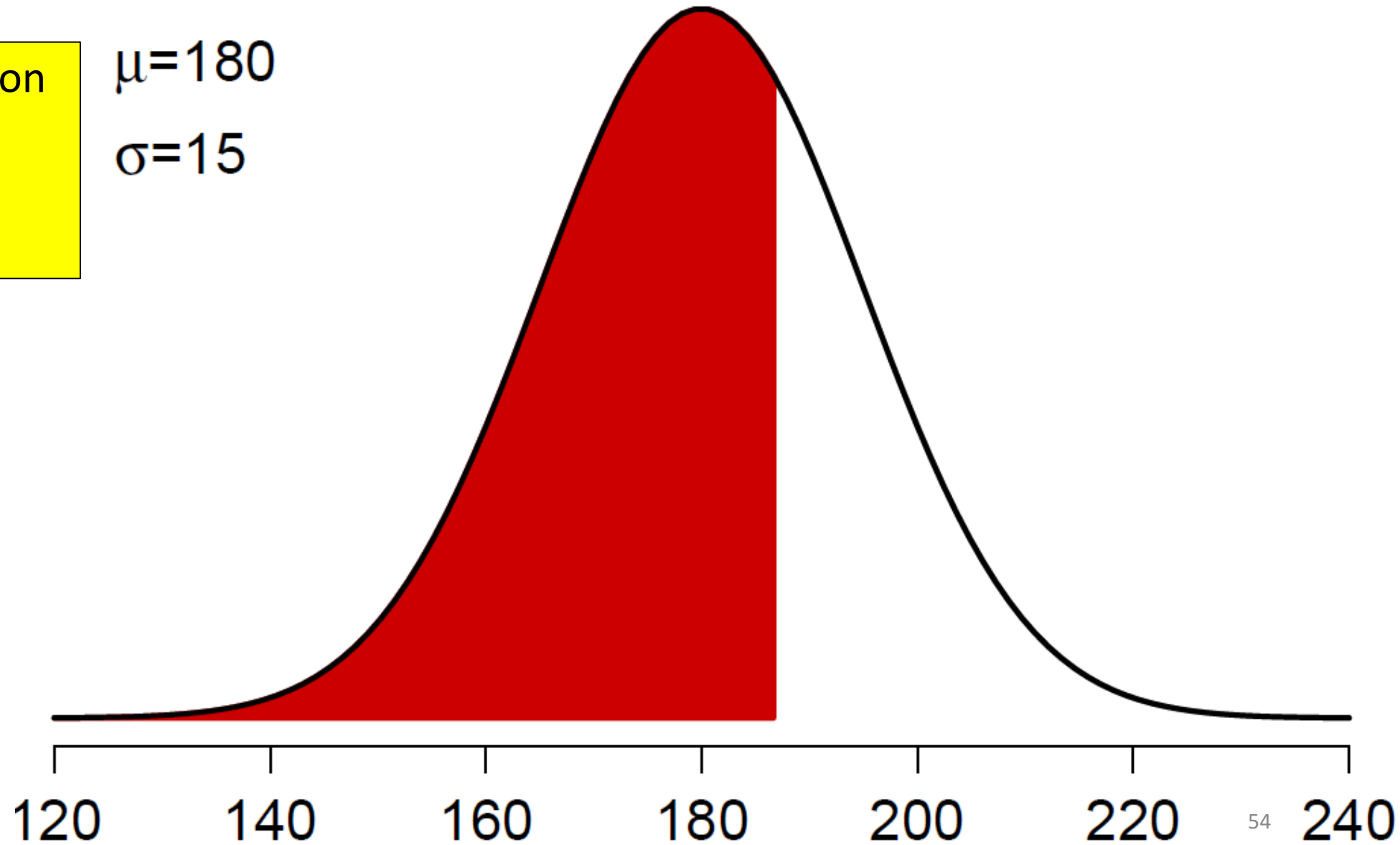
The normal distribution that is characterized by $\mu = 0$ and $\sigma = 1$ is called the 'standard normal distribution'

Step 1: Find z

Probability of a person
with a reaction time
less than 187.5?
i.e., $P(X < 187.5)$?

$$\mu = 180$$

$$\sigma = 15$$

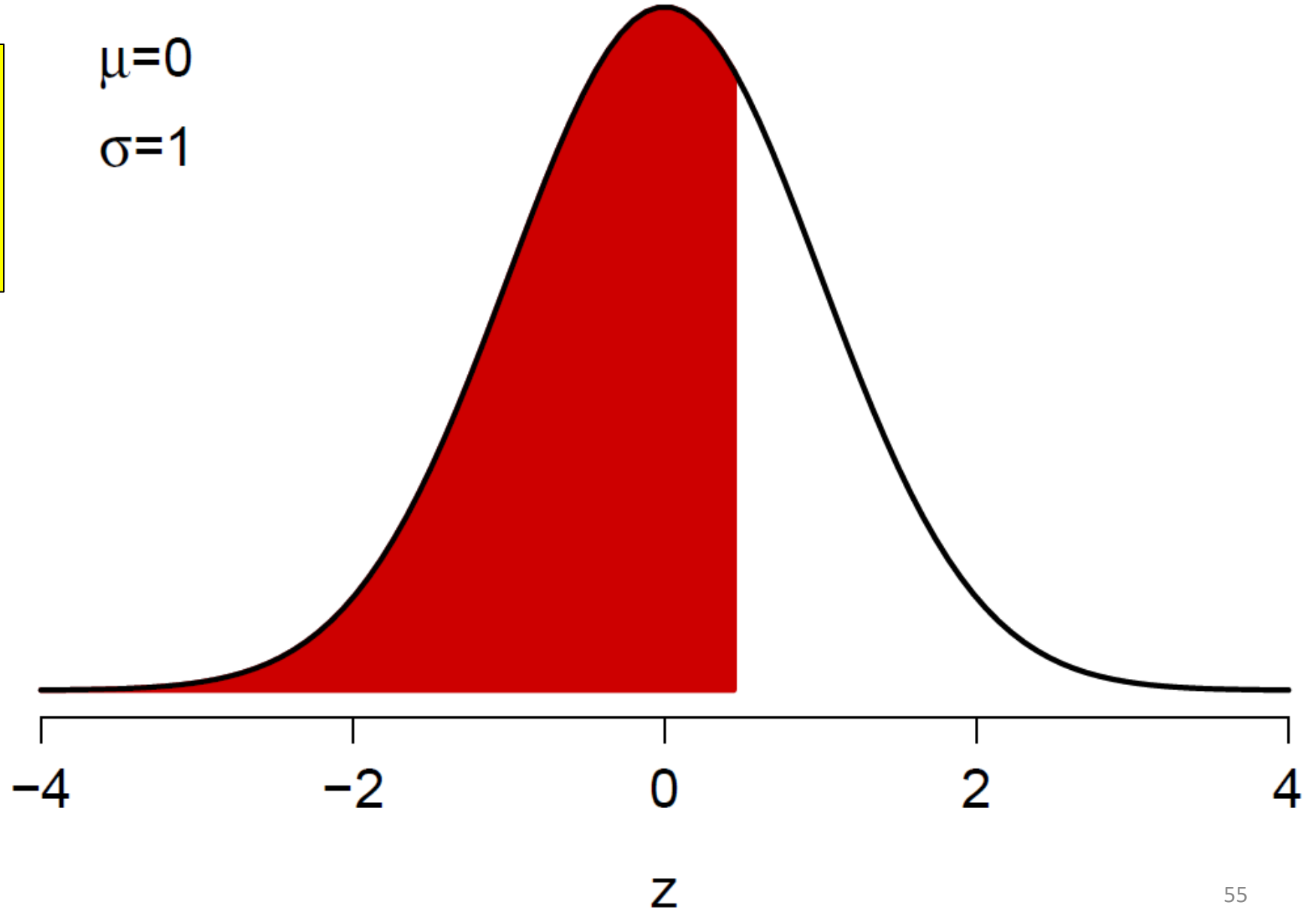


Step 2: Find the probability for z

Probability of a person
with a reaction time
less than 187.5?
i.e., $P(X < 187.5)$?

Corresponds to

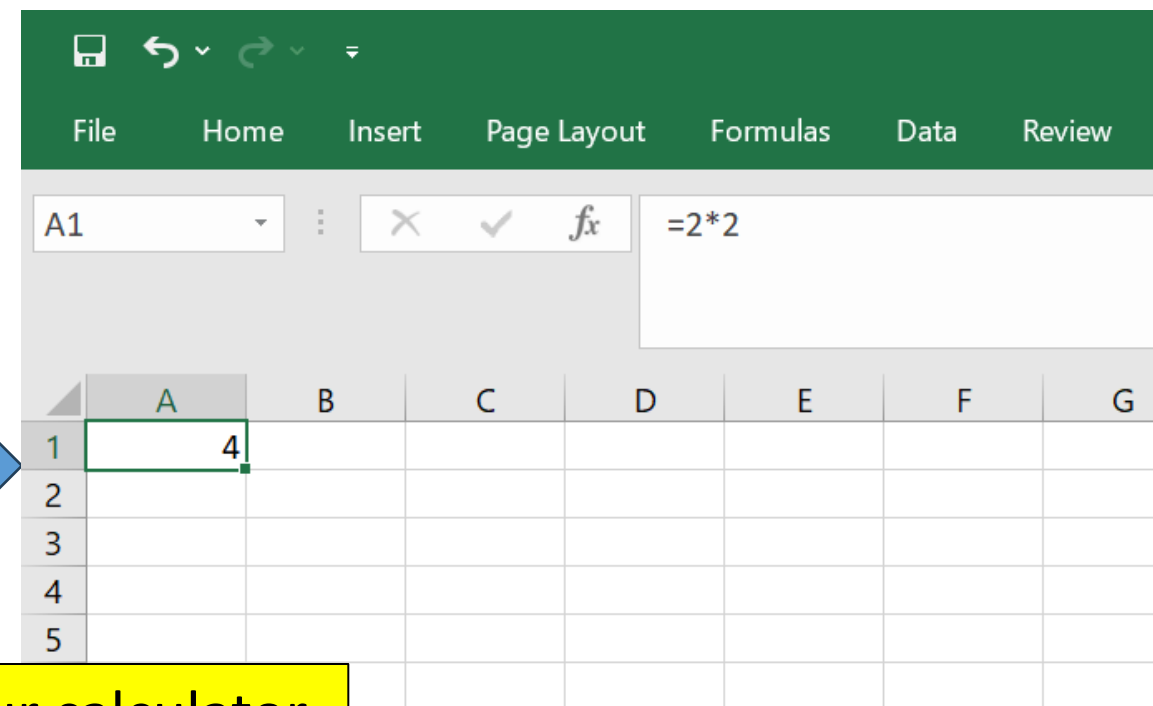
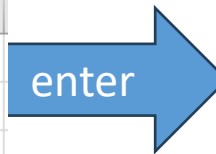
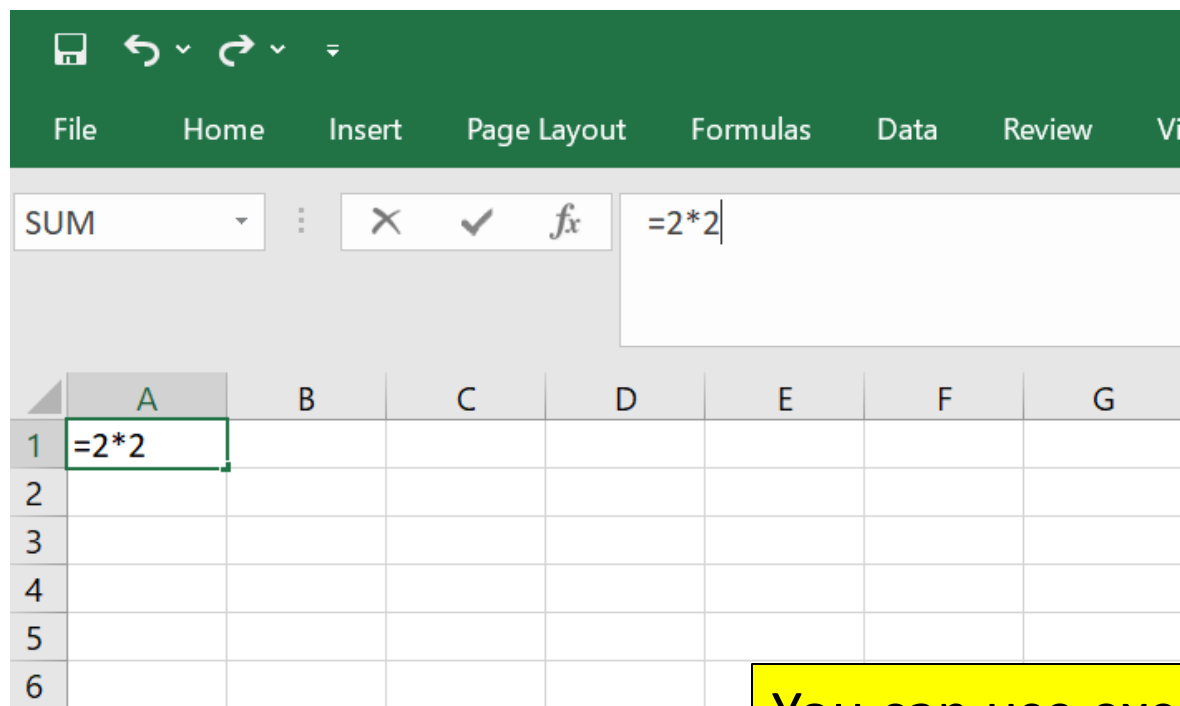
Probability of z
smaller than 0.5
i.e., $P(z < 0.5)$



Step 2: Find the probability for z

Welcome Excel!!

For more info on the use of Excel see:
Canvas -> Module -> Microsoft Excel
Some Excel commands will also be included in
formula sheet



You can use excel as your calculator

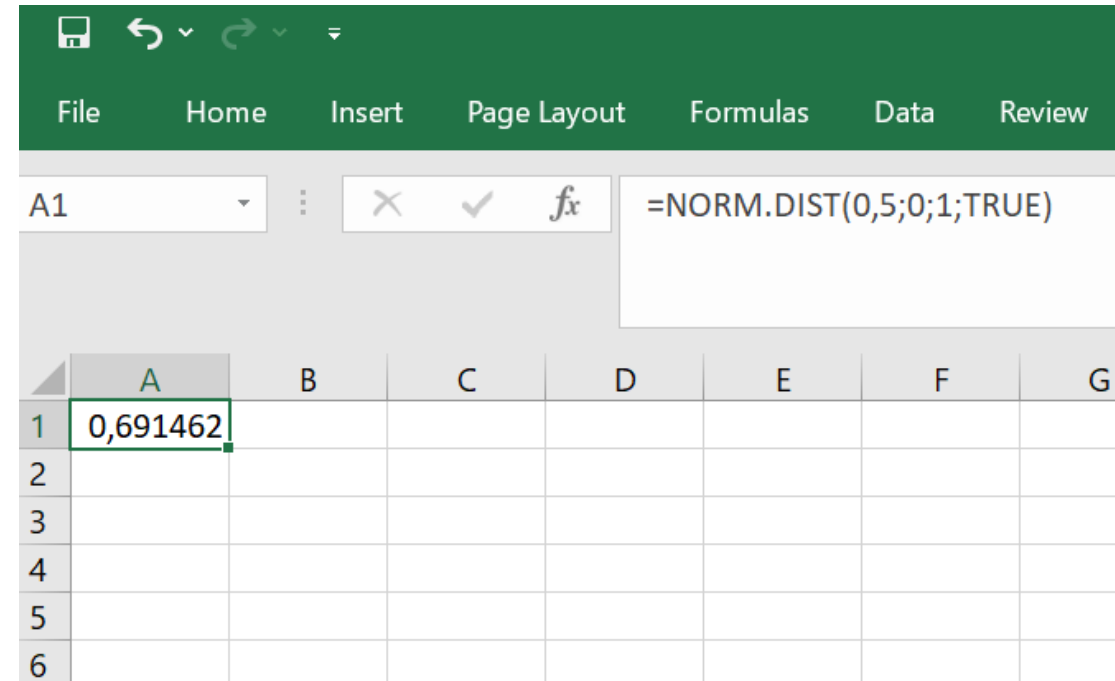
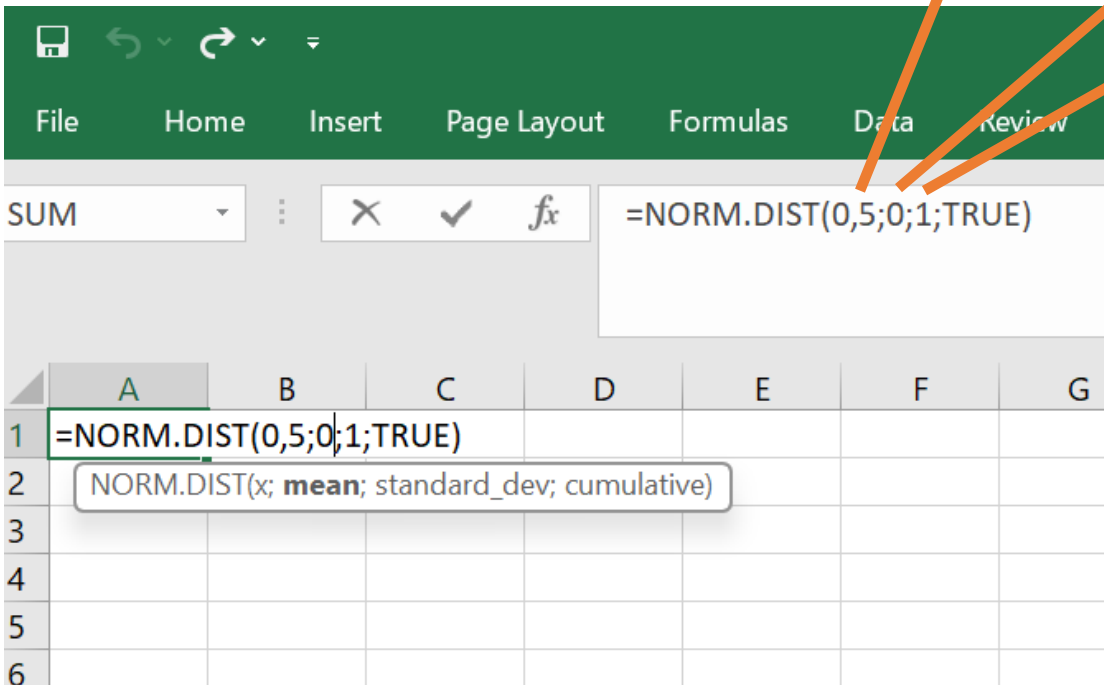
Step 2: Find the probability for z

What we want: Probability of z smaller than 0.5
i.e., $P(z < 0.5)$

Welcome Excel!!

= NORM.DIST(Z-value ; Mean = 0 ; SD = 1 ; TRUE)

Gives you probability mass LEFT of that z-value
(lower tail/smaller than z)



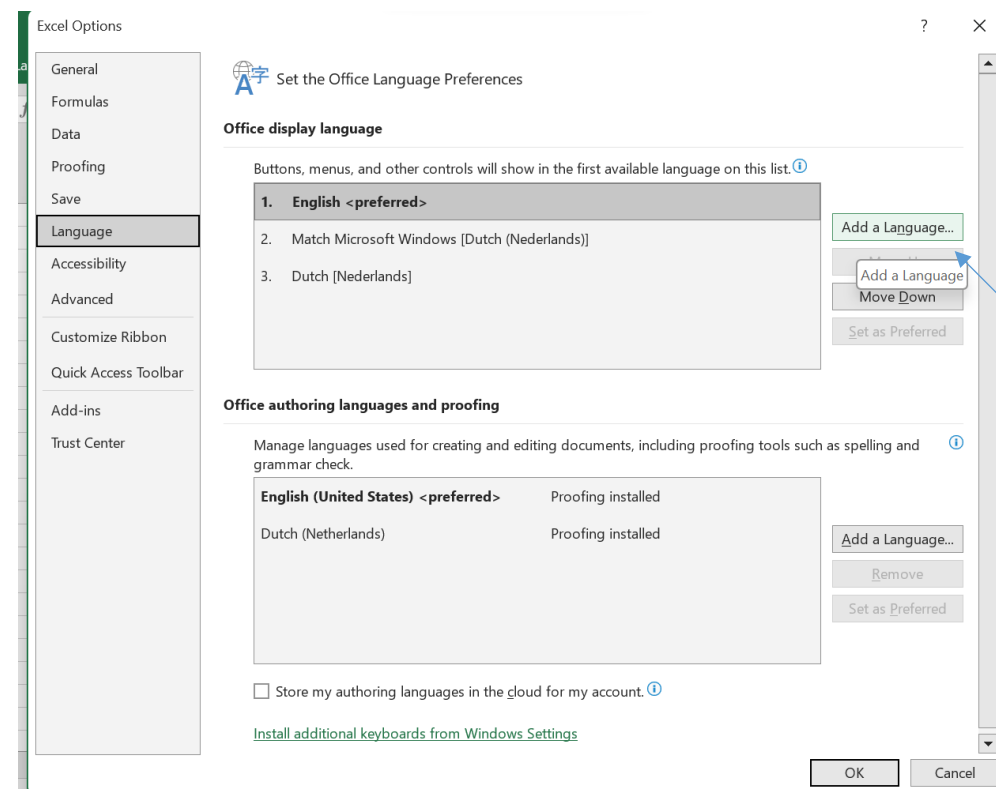
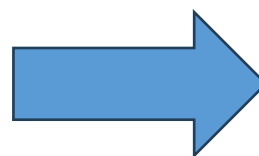
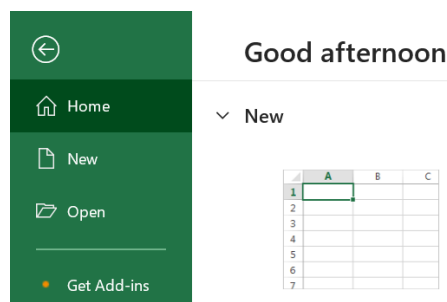
Step 2: Find the probability for z

What we want: Probability of z smaller than 0.5
i.e., $P(z < 0.5)$

Dutch Excel?

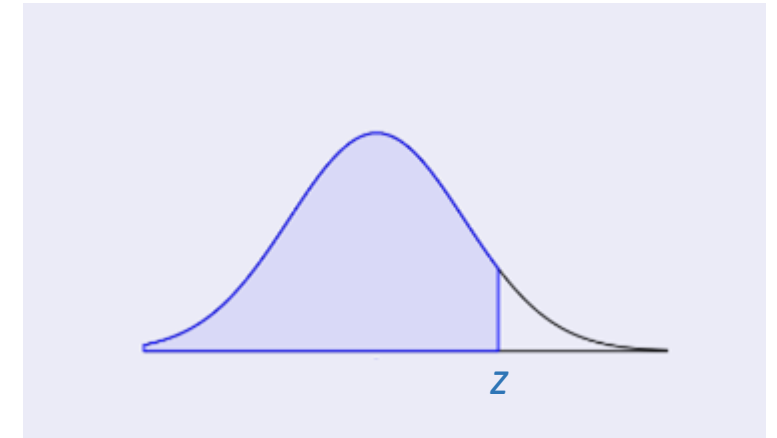
= NORM.VERD(Z-value ; Mean = 0 ; SD = 1 ; WAAR)

change the language:
Bestand > Opties > Taal



Step 2: Find the probability for z

When you are still practicing with excel,
you can check your answers with Appendix A of the book.
But, in the exam we expect you to use Excel!



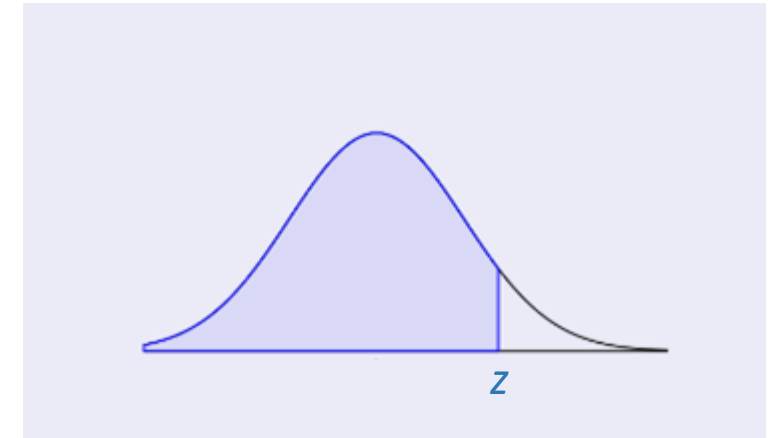
	.00	.01	.02	.03	.04	...
...	...	...	...	...	...	...
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	...
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	...
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	...
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	...
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	...
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	...
...	...	...	...	...	...	...

Probability of z
smaller than 0.5
i.e., $P(z < 0.5)$

So: $P(z < 0.5) = 0.6915 = P(\text{reaction time} < 178.5)$

Step 2: Find the probability for z

When you are still practicing with excel,
you can check your answers with Appendix A of the book.
But, in the exam we expect you to use Excel!



	.00	.01	.02	.03	.04	...
...	...	...	...	...	...	...
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	...
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	...
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	...
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	...
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	...
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	...
...	...	...	...	...	...	...

Probability of z
smaller than 0.53
i.e., $P(z < 0.53)$?

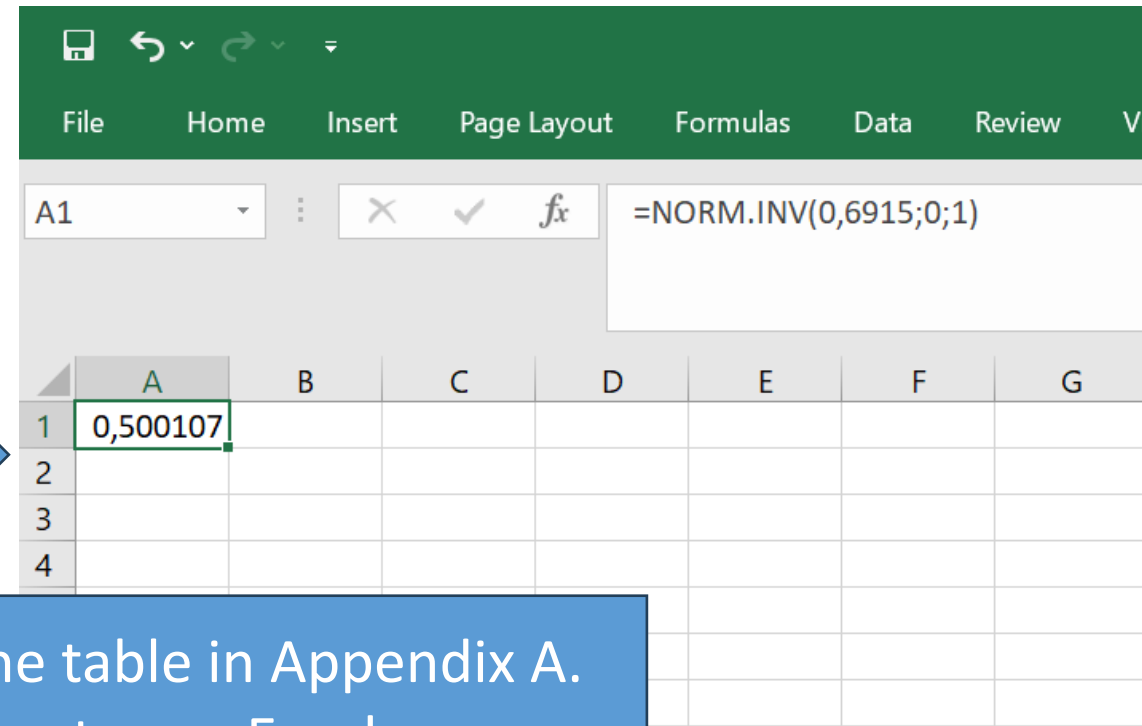
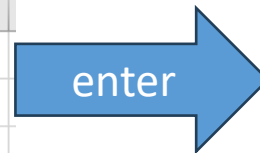
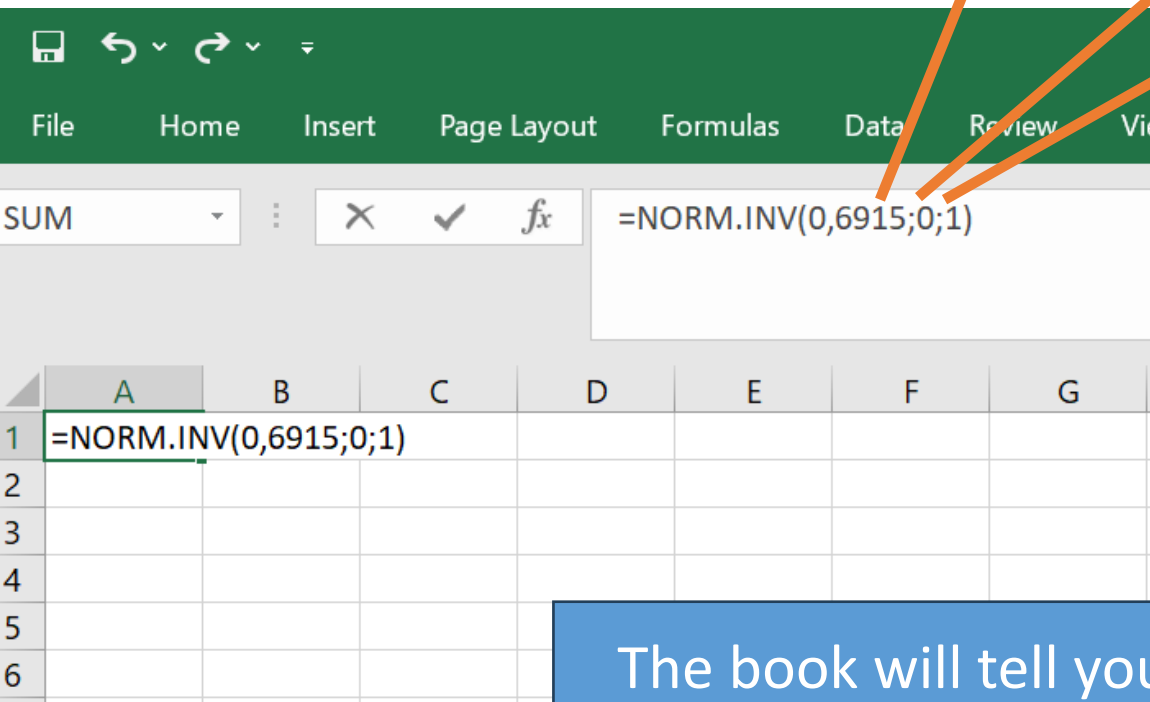
So: $P(z < 0.53) = 0.7019$

Inverse: Find the z for a probability

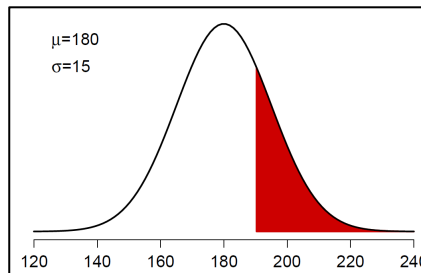
Suppose we want to know:
What value for z has a lower-tail probability of .6915 ?

Welcome Excel!!

= NORM.INV(probability ; Mean = 0 ; SD = 1)



The book will tell you to use the table in Appendix A.
However, we expect you to use Excel



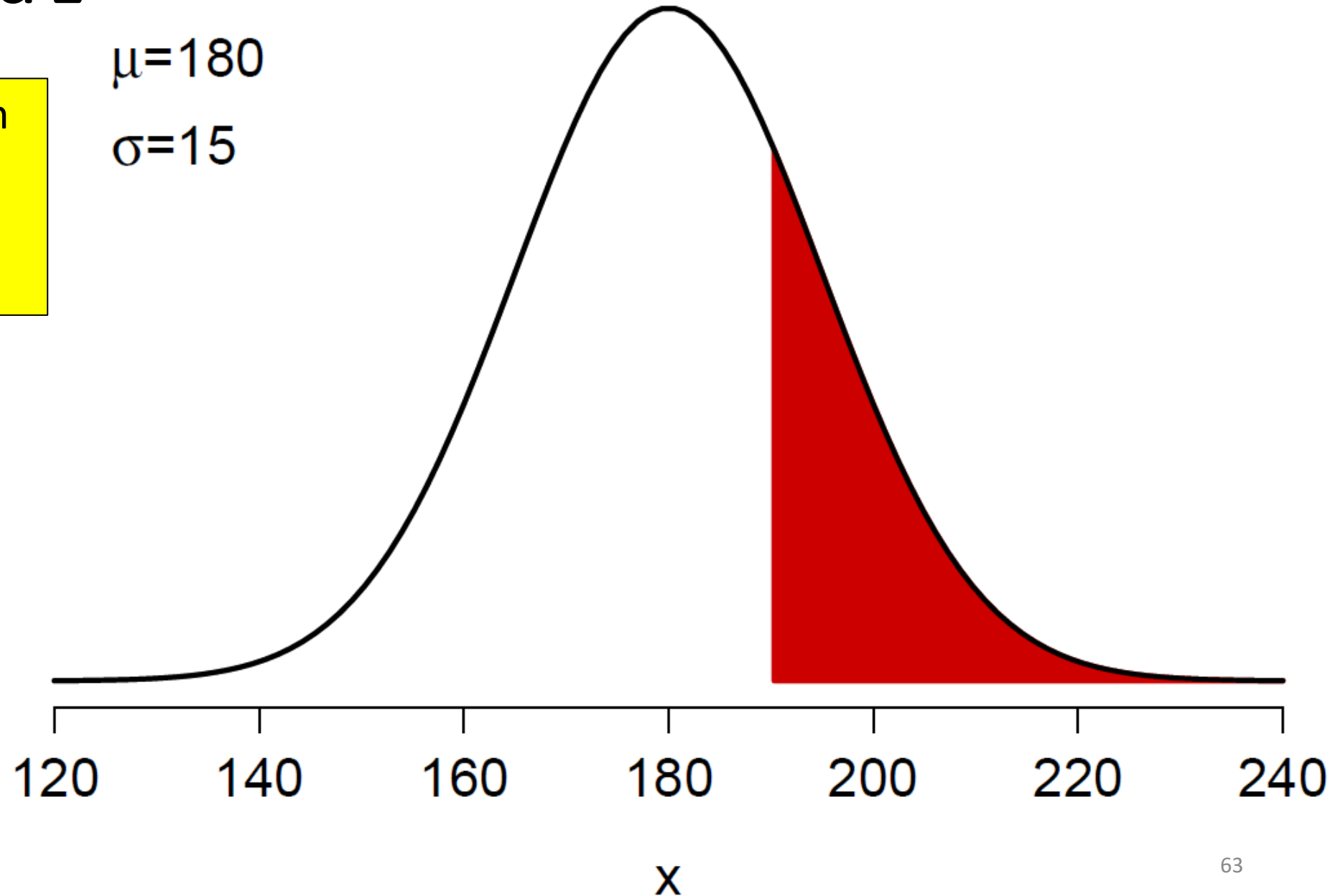
Situation II: Upper tail probability

Step 1: Find z

Probability of a person
with reaction time
longer than 190?
i.e., $P(X > 190)$?

$$\mu = 180$$
$$\sigma = 15$$

$$z = \frac{190 - 180}{15} = 0.67$$



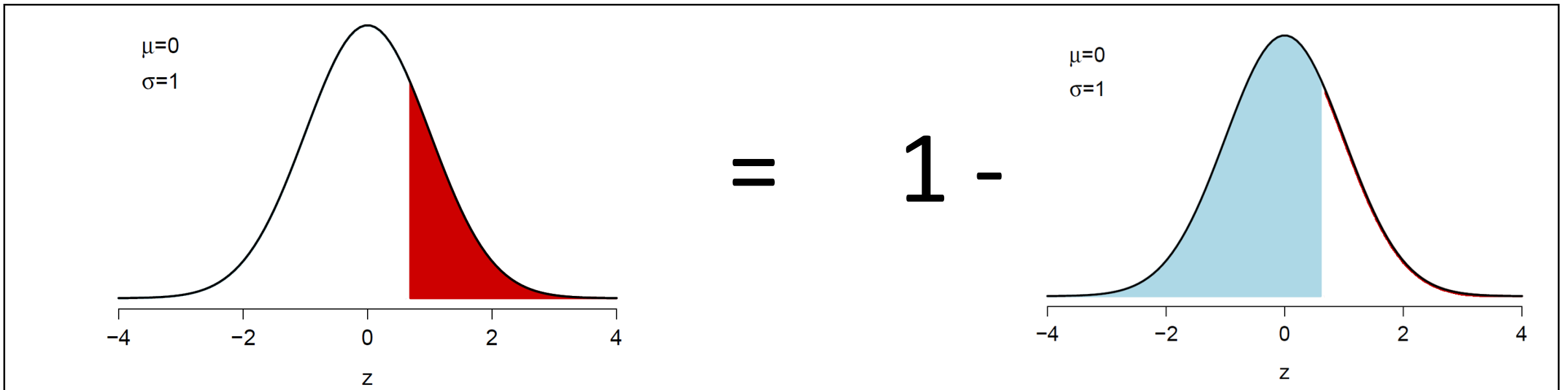
Step 2: Find the probability for z

Total probability is always 1:

- $P(z > 0.67) = 1 - P(z < 0.67)$

So: $P(z > 0.67) = 1 - P(z < 0.67) = 1 - 0.7486 = 0.2514$

File Home Insert Page Layout Formulas Data Review View							
A1		fx		=1-NORM.DIST(0,67;0;1;TRUE)			
	A	B	C	D	E	F	G
1	0,251429						
2							



Step 2: Find the probability for z

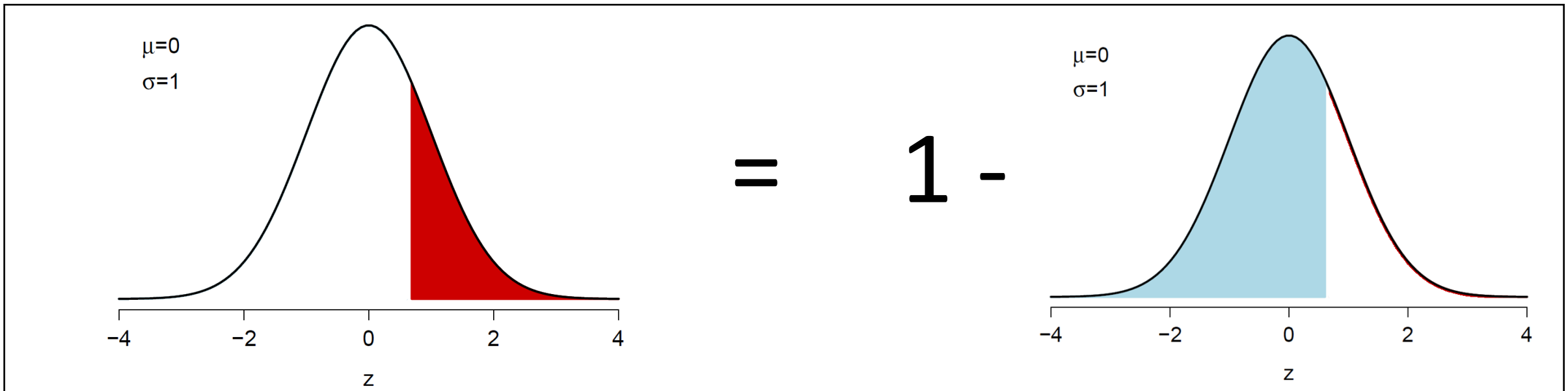
Total probability is always 1:

- $P(z > 0.67) = 1 - P(z < 0.67)$

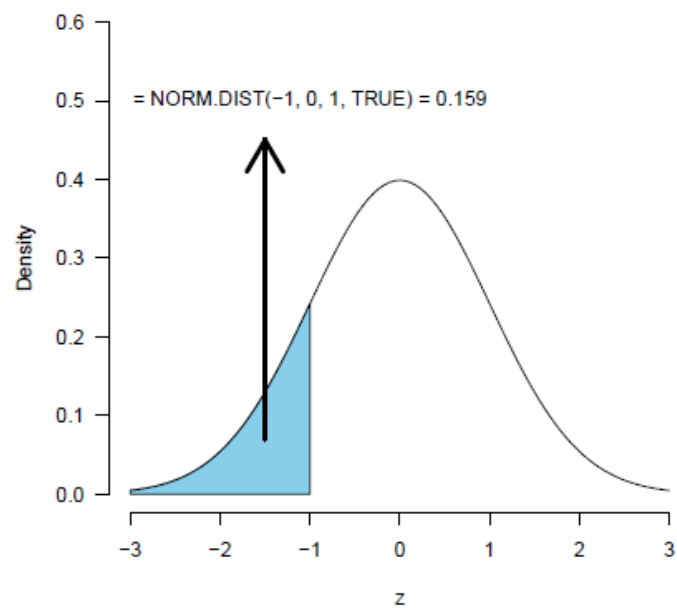
So: $P(z > 0.67) = 1 - P(z < 0.67) = 1 - 0.7486 = 0.2514$

You can check with Appendix A

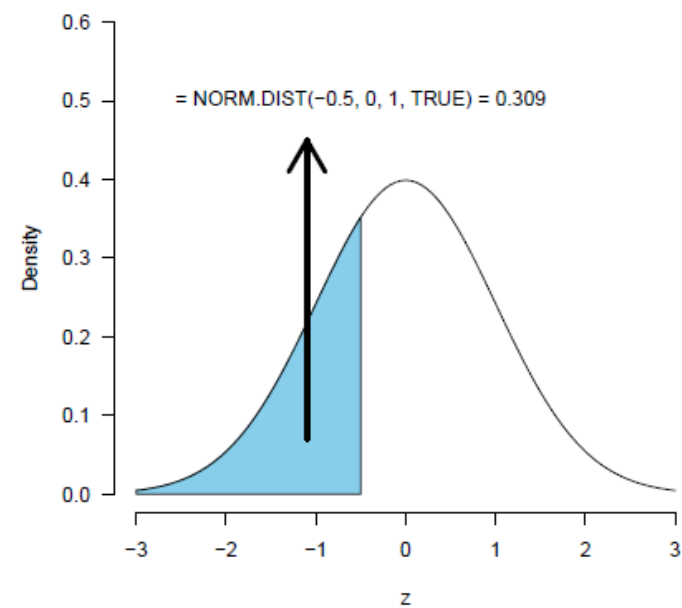
	...	0.05	0.06	0.07	0.08	0.09
...	...	...	...	...	...	...
0.3	...	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	...	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	...	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	...	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	...	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	...	0.8023	0.8051	0.8078	0.8106	0.8133
...	...	...	...	...	...	...



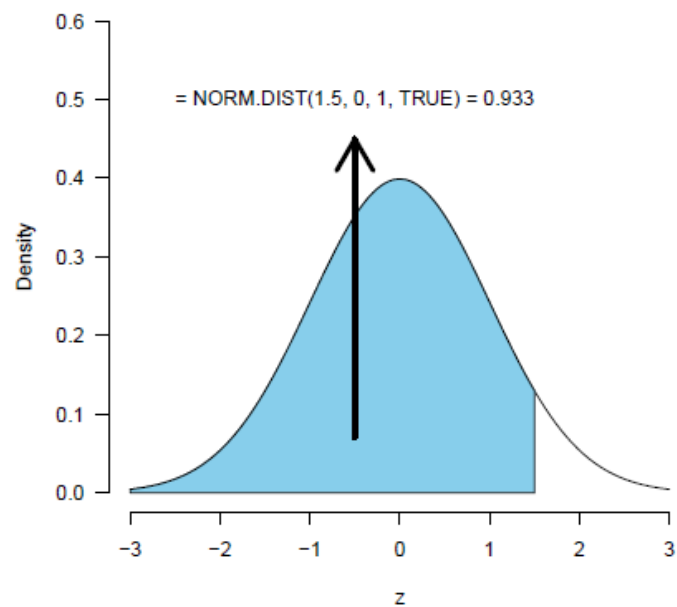
Standard normal distribution (z-distribution)



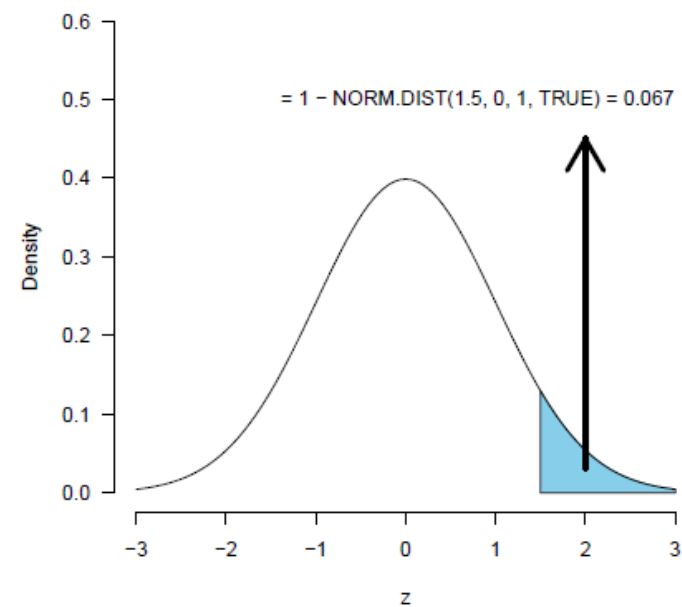
Standard normal distribution (z-distribution)



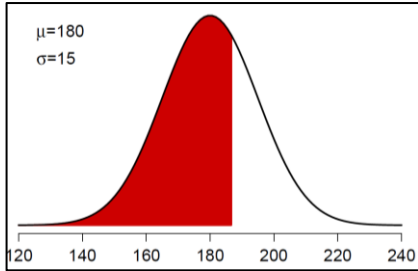
Standard normal distribution (z-distribution)



Standard normal distribution (z-distribution)

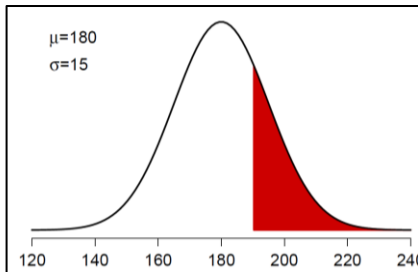


Three different probability calculations



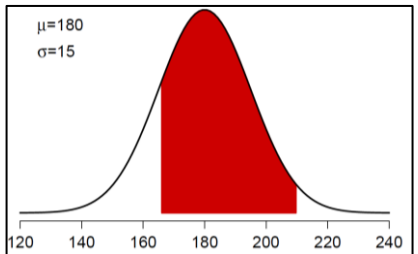
1) Probability of reaction time shorter than 187.5

- $P(X < 187.5)$



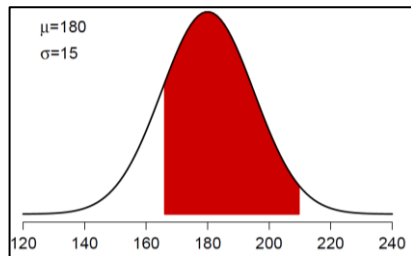
2) Probability of reaction time longer than 190

- $P(X > 190)$



3) Probability of reaction time between 165 and 210

- $P(165 < X < 210)$



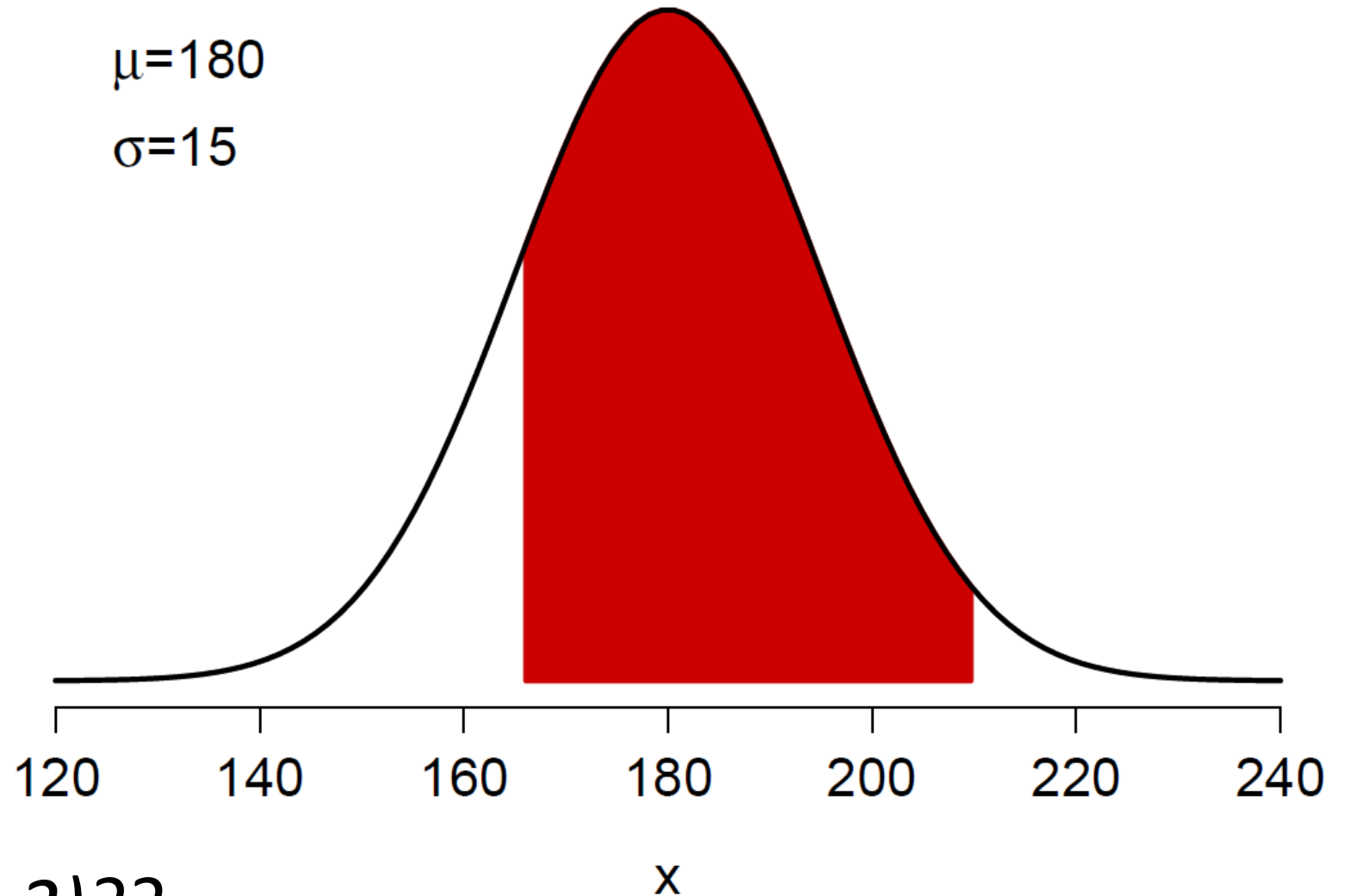
Situation III: Interval probability

Step 1: Find z

Probability of a person
with reaction time between
165 and 210 ?
i.e., $P(165 < X < 210)$?

$$z = \frac{165 - 180}{15} = -1$$

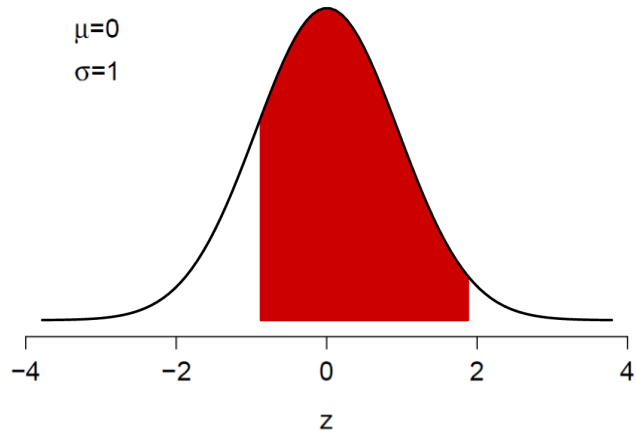
$$z = \frac{210 - 180}{15} = 2$$



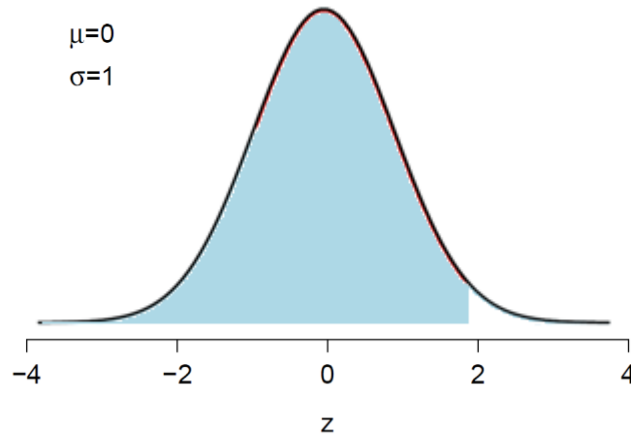
i.e., what is $P(-1 < z < 2)$??

Step 2: Find the probability for z

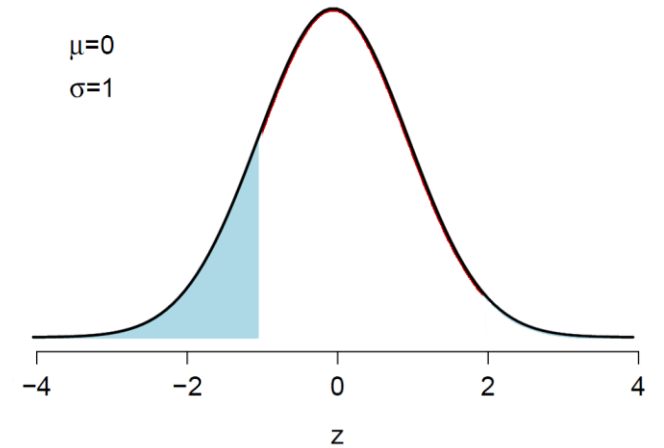
- $P(-1 < z < 2) = P(z < 2) - P(z < -1)$

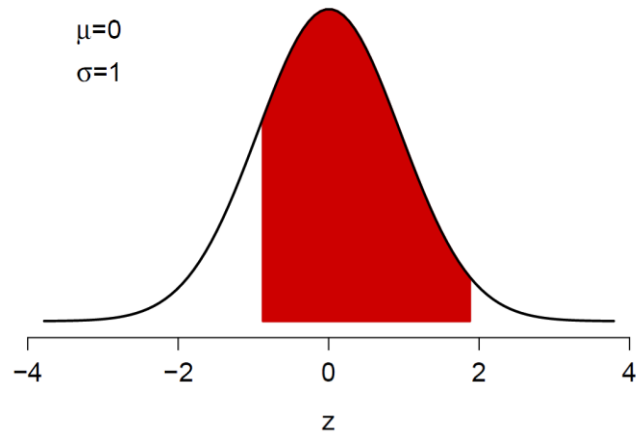


=

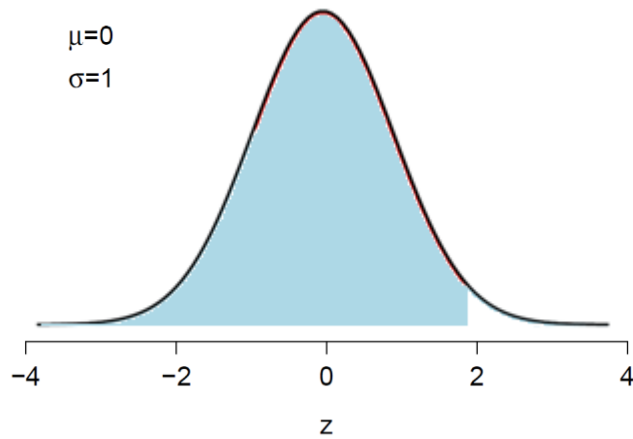


-

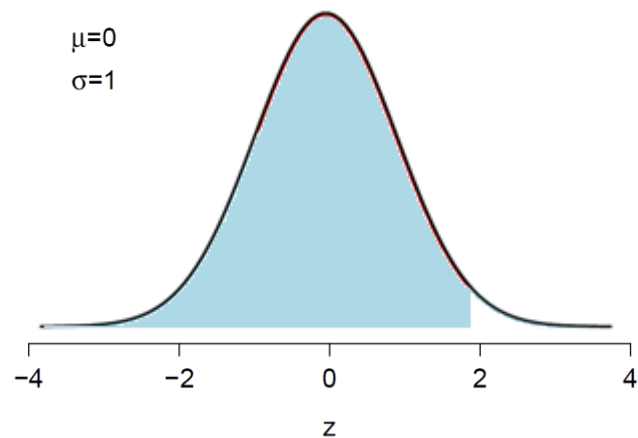
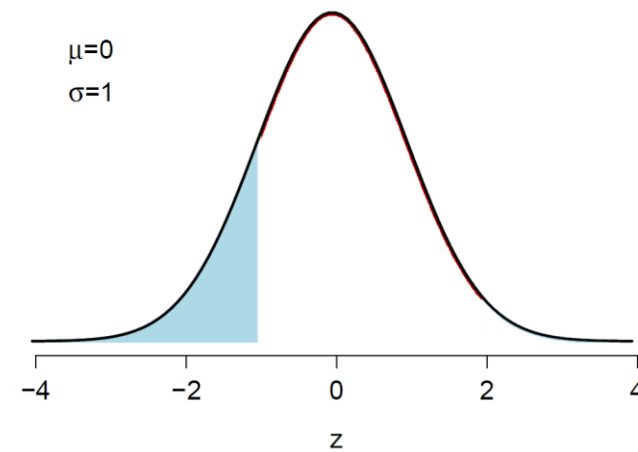




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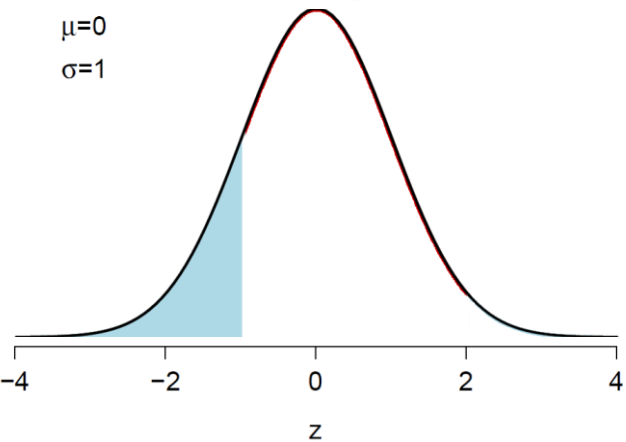


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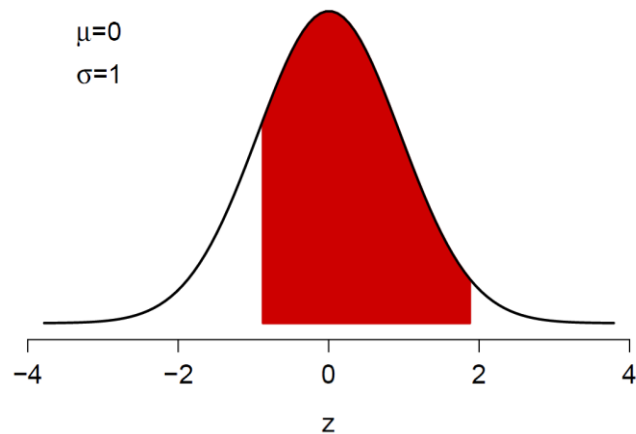
$$z = \frac{210 - 180}{15} = 2$$

$$P(z < 2) = 0.9772$$

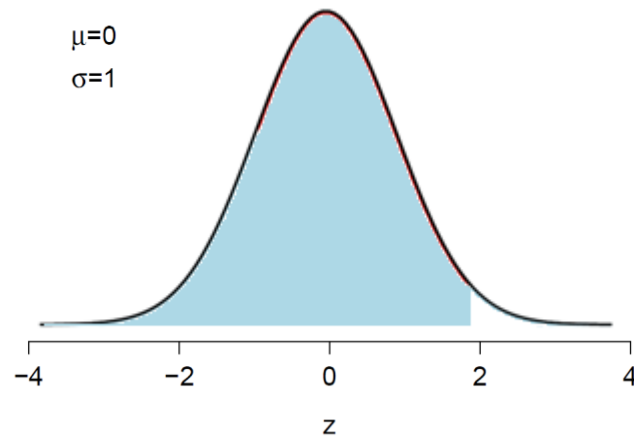


$$z = \frac{165 - 180}{15} = -1$$

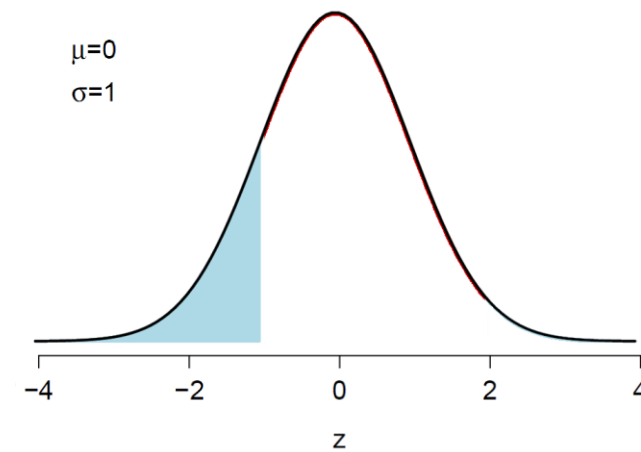
$$P(z < -1) = 0.1587$$



=



-



$$P(-1 < z < 2) = 0.9772 - 0.1587$$

$$= 0.8185$$

So: Probability of sampling someone with reaction time between 165 and 210 is 0.82

Today

- Random variables
- Probability distributions
 - Discrete variables
 - Three different probability calculations
 - Mean and variability
 - Continuous variables

Two specific distributions:

- Normal distribution (possible for continuous variables)
 - Mean and variability
 - Three different probability calculations
- Binomial distribution (possible for discrete variables)
 - **General formulae**
 - Mean and variability

Binomial distribution

- A probability distribution for a specific discrete variable
 - The specific discrete variable: number of 'successes' in n independent trials
 - This discrete variable is called the *random binomial variable*
 - Each trial has binary outcome (e.g., 'correct' & 'incorrect', or 'heads' & 'tails')
-
- Let's consider an example!

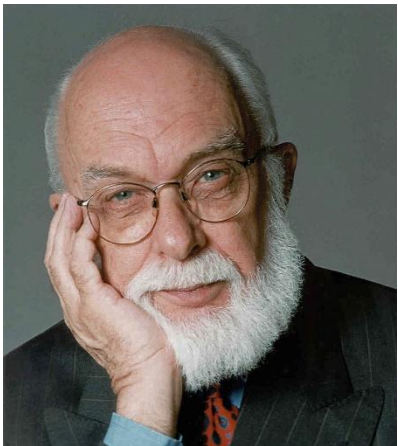


One-million dollar paranormal challenge by James Randi



Derek Ogilvie

https://nl.wikipedia.org/wiki/Derek_Ogilvie



James Randi

Picture: https://nl.wikipedia.org/wiki/James_Randi

Year	Challenger	Purported Ability	Test	Results	Notes
2007	Derek Ogilvie	Mediumship	Identify which one out of ten toys is being used by a child at a particular time.	Failed	
2009	Connie Sonne	Dowsing (Pendulum)	Identify playing cards in sealed envelope.	Failed ^[30]	
2010	Anita Ikonen	Medical dowsing	Determine by observation which of five subjects was missing a kidney.	Failed	Billed as "demonstration" not "test"
2011	No challenger available				
2012	Andrew Needles	Performance-enhancing bracelet	Distinguish participants wearing real product significant number of times.	Failed	
2013	Brahim Addoun	Remote viewing	Remotely identify 3 of 20 objects.	Failed	
2014	Fei Wang	Sending energy through his hand that can be felt by another person.	The energy should be felt by the target person correctly eight out of nine times.	Failed	
2015	No claimant – "Demonstration test"	Ethernet cables that are claimed to be "directional".	Volunteers were played sound twice and were asked to determine which cable had the highest sound quality.	Failed	

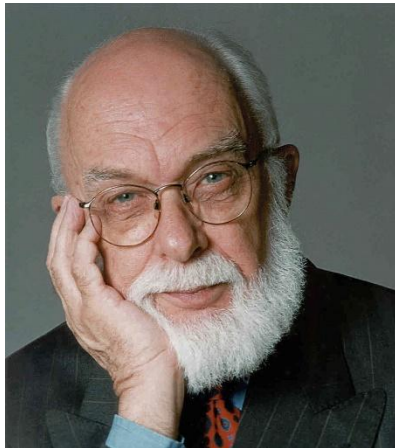
https://en.wikipedia.org/wiki/One_Million_Dollar_Paranormal_Challenge

Example: 1-Million dollar challenge by James Randi



Derek Ogilvie

https://nl.wikipedia.org/wiki/Derek_Ogilvie



James Randi

Picture: https://nl.wikipedia.org/wiki/James_Randi

- One trial has a **binary** outcome (correct / false):
- Let's pretend that Derek had 5 choices in each trial
 - If guessing, probability of correct = 0.2 (1 out of 5; $1/5$) and probability of incorrect = $1 - 0.2 = 0.8$
- And pretend that Derek had 4 trials
 - $n = 4$

outcome	#correct	probability
{C, C, C, C}	4	
{C, C, C, F}	3	

outcome	#correct	probability
{C, C, C, C}	4	
{C, C, C, F}	3	
{C, C, F, C}	3	
{C, F, C, C}	3	

outcome	#correct	probability
{C, C, C, C}	4	
{C, C, C, F}	3	
{C, C, F, C}	3	
{C, F, C, C}	3	
{F, C, C, C}	3	
{C, C, F, F}	2	
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

Recap from Chapter 5

→ Trials are independent

Thus: Product rule for independent events: $P(A \text{ and } B) = P(A) \times P(B)$

i.e., $P(\{\text{C}, \text{C}, \text{F}, \text{F}\}) = P(\text{C}) \times P(\text{C}) \times P(\text{F}) \times P(\text{F})$

$P(\text{C}) = 0.2$ (probability of a correct guess, 1 out of 5)

$P(\text{F}) = 1 - 0.2 = 0.8$ (complement rule)

$P(\{\text{C}, \text{C}, \text{F}, \text{F}\}) = P(\text{C}) \times P(\text{C}) \times P(\text{F}) \times P(\text{F}) = 0.2 \times 0.2 \times 0.8 \times 0.8 = 0.0256$

outcome	#correct	probability
{C, C, C, C}	4	
{C, C, C, F}	3	
{C, C, F, C}	3	
{C, F, C, C}	3	
{F, C, C, C}	3	
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	
{C, C, F, C}	3	
{C, F, C, C}	3	
{F, C, C, C}	3	
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	
{C, F, C, C}	3	
{F, C, C, C}	3	
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	
{F, C, C, C}	3	
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	
{C, F, C, F}	2	
{F, C, F, C}	2	
{F, F, C, C}	2	
{F, C, C, F}	2	
{C, F, F, F}	1	
{F, C, F, F}	1	
{F, F, C, F}	1	
{F, F, F, C}	1	
{F, F, F, F}	0	

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	$0.2 \times 0.8 \times 0.8 \times 0.2 = 0.0256$
{C, F, C, F}	2	$0.2 \times 0.8 \times 0.2 \times 0.8 = 0.0256$
{F, C, F, C}	2	$0.8 \times 0.2 \times 0.8 \times 0.2 = 0.0256$
{F, F, C, C}	2	$0.8 \times 0.8 \times 0.2 \times 0.2 = 0.0256$
{F, C, C, F}	2	$0.8 \times 0.2 \times 0.2 \times 0.8 = 0.0256$
{C, F, F, F}	1	$0.2 \times 0.8 \times 0.8 \times 0.8 = 0.1024$
{F, C, F, F}	1	$0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$
{F, F, C, F}	1	$0.8 \times 0.8 \times 0.2 \times 0.8 = 0.1024$
{F, F, F, C}	1	$0.8 \times 0.8 \times 0.8 \times 0.2 = 0.1024$
{F, F, F, F}	0	$0.8 \times 0.8 \times 0.8 \times 0.8 = 0.4096$

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	$0.2 \times 0.8 \times 0.8 \times 0.2 = 0.0256$
{C, F, C, F}	2	$0.2 \times 0.8 \times 0.2 \times 0.8 = 0.0256$
{F, C, F, C}	2	$0.8 \times 0.2 \times 0.8 \times 0.2 = 0.0256$
{F, F, C, C}	2	$0.8 \times 0.8 \times 0.2 \times 0.2 = 0.0256$
{F, C, C, F}	2	$0.8 \times 0.2 \times 0.2 \times 0.8 = 0.0256$
{C, F, F, F}	1	$0.2 \times 0.8 \times 0.8 \times 0.8 = 0.1024$
{F, C, F, F}	1	$0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$
{F, F, C, F}	1	$0.8 \times 0.8 \times 0.2 \times 0.8 = 0.1024$
{F, F, F, C}	1	$0.8 \times 0.8 \times 0.8 \times 0.2 = 0.1024$
{F, F, F, F}	0	$0.8 \times 0.8 \times 0.8 \times 0.8 = 0.4096$

X: Number correct

What is the probability of 3 correct?

- $P(X = 3)$
- $= P(\{C, C, C, F\} \text{ OR } \{C, C, F, C\} \text{ OR } \{C, F, C, C\} \text{ OR } \{F, C, C, C\})$

$$= 0.0064 + 0.0064 + 0.0064 + 0.0064$$

$$= 4 \times 0.0064 = 0.0256$$

Addition rule for disjoint events

Number of ways you can guess 3 items correctly

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	$0.2 \times 0.8 \times 0.8 \times 0.2 = 0.0256$
{C, F, C, F}	2	$0.2 \times 0.8 \times 0.2 \times 0.8 = 0.0256$
{F, C, F, C}	2	$0.8 \times 0.2 \times 0.8 \times 0.2 = 0.0256$
{F, F, C, C}	2	$0.8 \times 0.8 \times 0.2 \times 0.2 = 0.0256$
{F, C, C, F}	2	$0.8 \times 0.2 \times 0.2 \times 0.8 = 0.0256$
{C, F, F, F}	1	$0.2 \times 0.8 \times 0.8 \times 0.8 = 0.1024$
{F, C, F, F}	1	$0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$
{F, F, C, F}	1	$0.8 \times 0.8 \times 0.2 \times 0.8 = 0.1024$
{F, F, F, C}	1	$0.8 \times 0.8 \times 0.8 \times 0.2 = 0.1024$
{F, F, F, F}	0	$0.8 \times 0.8 \times 0.8 \times 0.8 = 0.4096$

X: Number correct

What is the probability of 2 correct?

- $P(X = 2)$
- $= 0.0256 + 0.0256 + 0.0256 + 0.0256 + 0.0256 + 0.0256$

$$= 6 \times 0.0256 = 0.1536$$

Number of ways you can guess 2 items correctly

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	$0.2 \times 0.8 \times 0.8 \times 0.2 = 0.0256$
{C, F, C, F}	2	$0.2 \times 0.8 \times 0.2 \times 0.8 = 0.0256$
{F, C, F, C}	2	$0.8 \times 0.2 \times 0.8 \times 0.2 = 0.0256$
{F, F, C, C}	2	$0.8 \times 0.8 \times 0.2 \times 0.2 = 0.0256$
{F, C, C, F}	2	$0.8 \times 0.2 \times 0.2 \times 0.8 = 0.0256$
{C, F, F, F}	1	$0.2 \times 0.8 \times 0.8 \times 0.8 = 0.1024$
{F, C, F, F}	1	$0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$
{F, F, C, F}	1	$0.8 \times 0.8 \times 0.2 \times 0.8 = 0.1024$
{F, F, F, C}	1	$0.8 \times 0.8 \times 0.8 \times 0.2 = 0.1024$
{F, F, F, F}	0	$0.8 \times 0.8 \times 0.8 \times 0.8 = 0.4096$

General formula

- We have n trials of which the outcome is binary ('success' and 'no success')
- The probability of success in a trial is p
- The probability of no success in a trial is $1 - p$

Factorial in excel:
 $x! = \text{fact}(x)$

Then: the probability of x successes is given by:

$$P(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

! is the factorial function:
 $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
 $3! = 3 \times 2 \times 1 = 6$
 $1! = 1$
 $0! = 1$

Number of ways in which you get x successes in n trials

Probability

How many permutations (ways to order) are there of 0 numbers?
One! (doing nothing)

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	$0.2 \times 0.8 \times 0.8 \times 0.2 = 0.0256$
{C, F, C, F}	2	$0.2 \times 0.8 \times 0.2 \times 0.8 = 0.0256$
{F, C, F, C}	2	$0.8 \times 0.2 \times 0.8 \times 0.2 = 0.0256$
{F, F, C, C}	2	$0.8 \times 0.8 \times 0.2 \times 0.2 = 0.0256$
{F, C, C, F}	2	$0.8 \times 0.2 \times 0.2 \times 0.8 = 0.0256$
{C, F, F, F}	1	$0.2 \times 0.8 \times 0.8 \times 0.8 = 0.1024$
{F, C, F, F}	1	$0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$
{F, F, C, F}	1	$0.8 \times 0.8 \times 0.2 \times 0.8 = 0.1024$
{F, F, F, C}	1	$0.8 \times 0.8 \times 0.8 \times 0.2 = 0.1024$
{F, F, F, F}	0	$0.8 \times 0.8 \times 0.8 \times 0.8 = 0.4096$

General formula

- We have n trials of which the outcome is binary ('success' and 'no success')
- The probability of success in a trial is p
- The probability of no success in a trial is $1 - p$

Then: the probability of x successes is given by:

$$P(x) = \frac{n!}{x! (n - x)!} p^x (1 - p)^{n-x}$$

For this example about Ogilvie

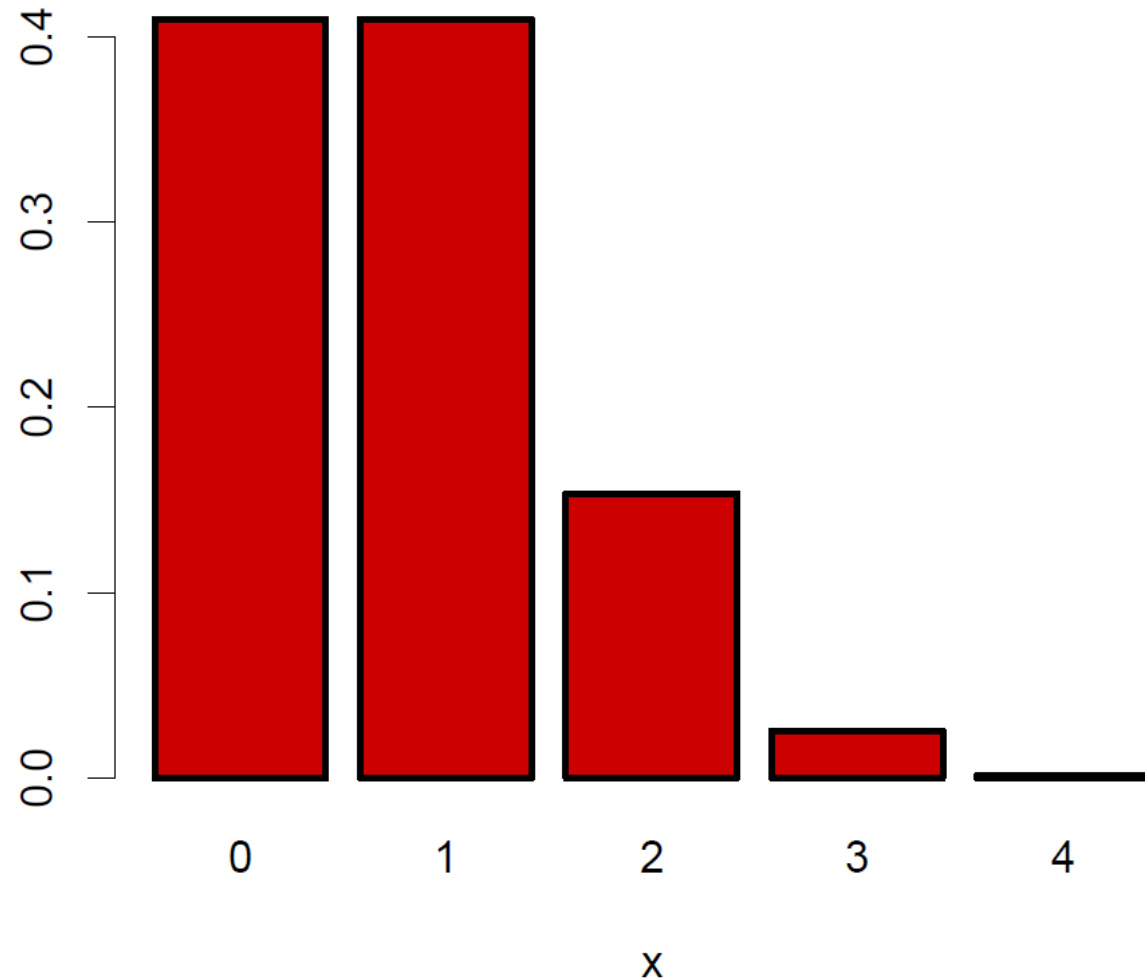
- $P(4) = 1 \times 0.20^4 \times 0.80^0 = 0.0016$
- $P(3) = 4 \times 0.20^3 \times 0.80^1 = 0.0256$
- $P(2) = 6 \times 0.20^2 \times 0.80^2 = 0.1536$
- $P(1) = 4 \times 0.20^1 \times 0.80^3 = 0.4096$
- $P(0) = 1 \times 0.20^0 \times 0.80^4 = 0.4096$

So, if Derek is guessing, with 4 trials and $p=0.2$, then this is what we can expect:



Derek Ogilvie

https://nl.wikipedia.org/wiki/Derek_Ogilvie



Today

- Random variables
- Probability distributions
 - Discrete variables
 - Three different probability calculations
 - Mean and variability
 - Continuous variables

Two specific distributions:

- Normal distribution (possible for continuous variables)
 - Mean and variability
 - Three different probability calculations
- Binomial distribution (possible for discrete variables)
 - General formulae
 - **Mean and variability**

outcome	#correct	probability
{C, C, C, C}	4	$0.2 \times 0.2 \times 0.2 \times 0.2 = 0.0016$
{C, C, C, F}	3	$0.2 \times 0.2 \times 0.2 \times 0.8 = 0.0064$
{C, C, F, C}	3	$0.2 \times 0.2 \times 0.8 \times 0.2 = 0.0064$
{C, F, C, C}	3	$0.2 \times 0.8 \times 0.2 \times 0.2 = 0.0064$
{F, C, C, C}	3	$0.8 \times 0.2 \times 0.2 \times 0.2 = 0.0064$
{C, C, F, F}	2	0.0256 (see previous slide)
{C, F, F, C}	2	$0.2 \times 0.8 \times 0.8 \times 0.2 = 0.0256$
{C, F, C, F}	2	$0.2 \times 0.8 \times 0.2 \times 0.8 = 0.0256$
{F, C, F, C}	2	$0.8 \times 0.2 \times 0.8 \times 0.2 = 0.0256$
{F, F, C, C}	2	$0.8 \times 0.8 \times 0.2 \times 0.2 = 0.0256$
{F, C, C, F}	2	$0.8 \times 0.2 \times 0.2 \times 0.8 = 0.0256$
{C, F, F, F}	1	$0.2 \times 0.8 \times 0.8 \times 0.8 = 0.1024$
{F, C, F, F}	1	$0.8 \times 0.2 \times 0.8 \times 0.8 = 0.1024$
{F, F, C, F}	1	$0.8 \times 0.8 \times 0.2 \times 0.8 = 0.1024$
{F, F, F, C}	1	$0.8 \times 0.8 \times 0.8 \times 0.2 = 0.1024$
{F, F, F, F}	0	$0.8 \times 0.8 \times 0.8 \times 0.8 = 0.4096$

Mean:

- For the binomial, the more general formula for the mean of discrete variables ($\mu = \sum xP(x)$) simplifies to :

$$\mu = np$$

- With $n = 4$ trials $p = 0.2$ probability success, we had calculated:

- $P(4) = 1 \times 0.20^4 \times 0.80^0 = 0.0016$
- $P(3) = 4 \times 0.20^3 \times 0.80^1 = 0.0256$
- $P(2) = 6 \times 0.20^2 \times 0.80^2 = 0.1536$
- $P(1) = 4 \times 0.20^1 \times 0.80^3 = 0.4096$
- $P(0) = 1 \times 0.20^0 \times 0.80^4 = 0.4096$

$$\mu = \sum xP(x) = 0.4096 + (2 \times 0.1536) + (3 \times 0.0256) + (4 \times 0.0016) = 0.8$$

$$\mu = np = 4 \times 0.2 = 0.8$$

Standard deviation

$$\sigma = \sqrt{np(1 - p)}$$

If Derek guesses and $n=4$,
 $p=0.2$, then this is what we
can expect:

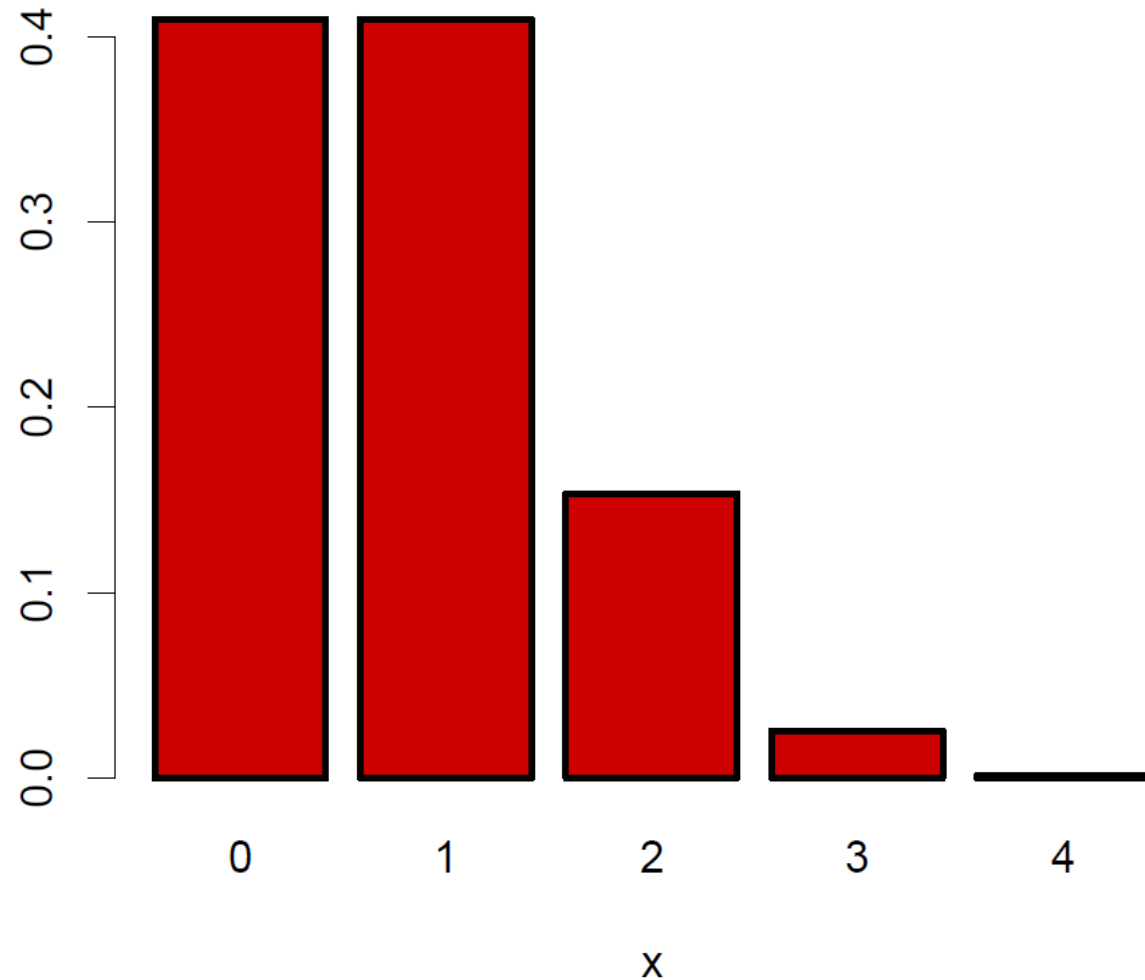


Derek Ogilvie

https://nl.wikipedia.org/wiki/Derek_Ogilvie

$$\mu = np = 4 \times 0.2 = 0.8$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{4 \times 0.2 \times (1-0.2)} = 0.8$$



Increasing the trials..
n=10, $p=0.2$, then this is
what we can expect:

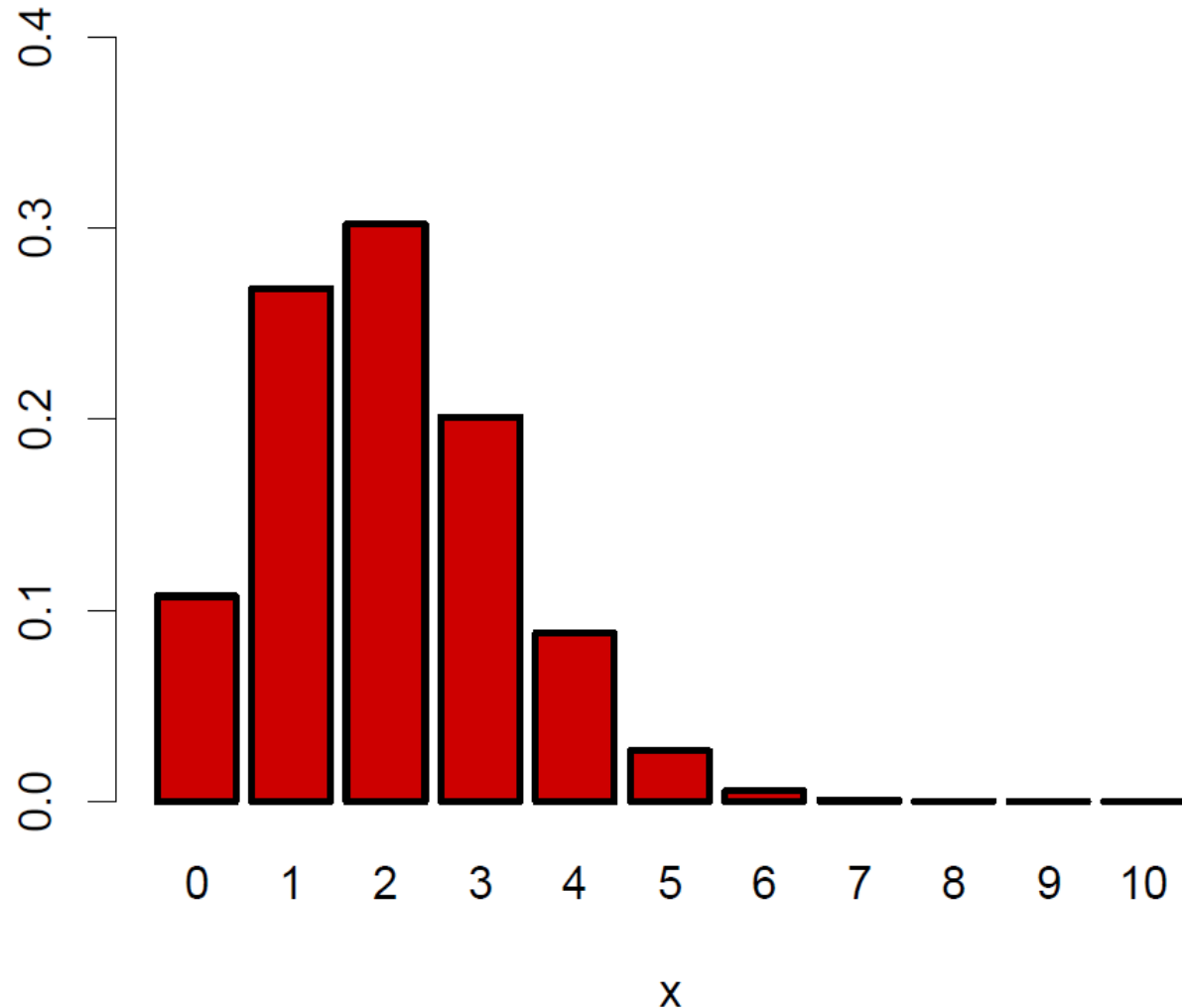


Derek Ogilvie

https://nl.wikipedia.org/wiki/Derek_Ogilvie

$$\mu = np = 10 \times 0.2 = 2$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{10 \times 0.2 \times (1-0.2)} = 1.26$$

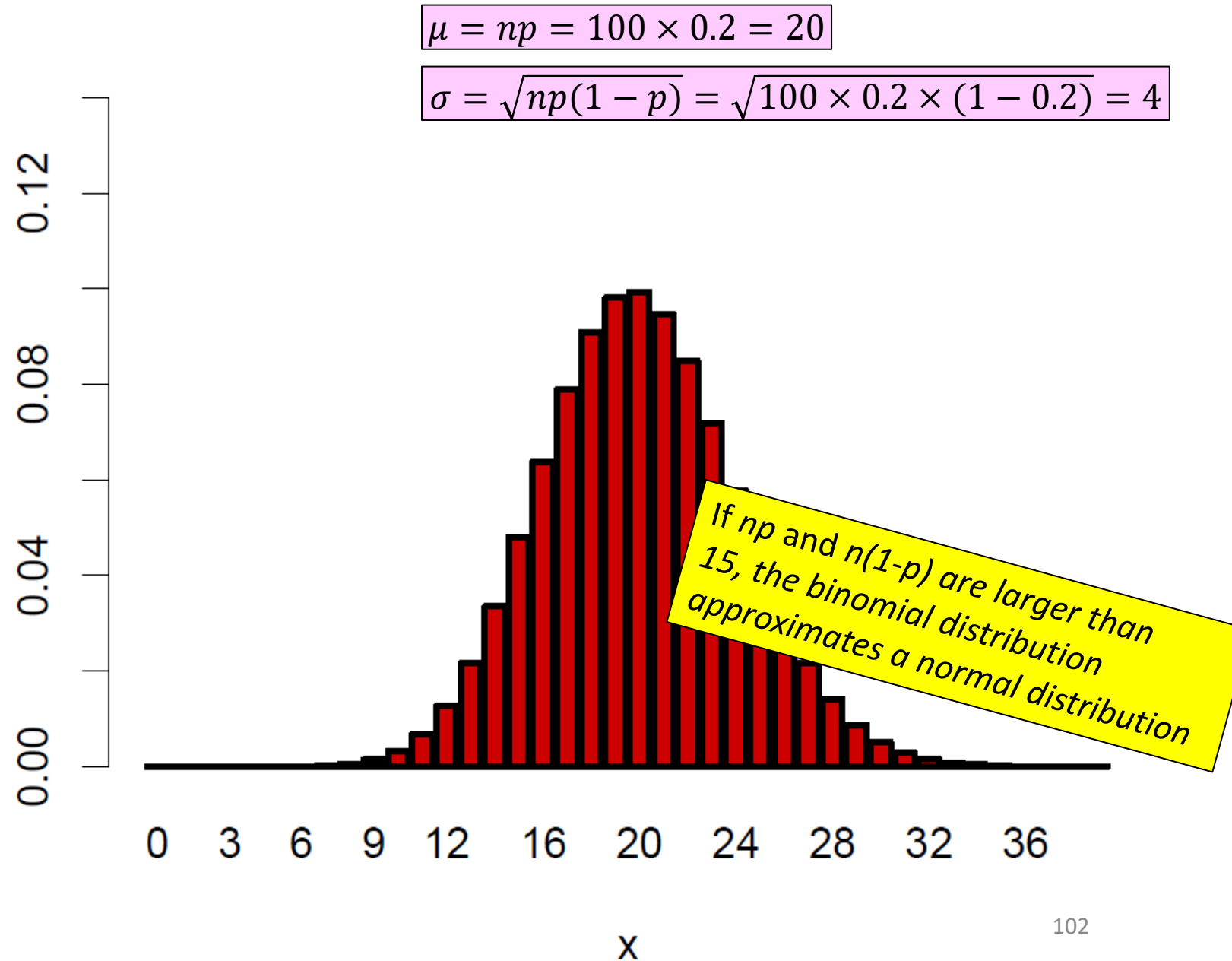


Increasing the trials..
n=100, $p=0.2$, then this is
what we can expect:



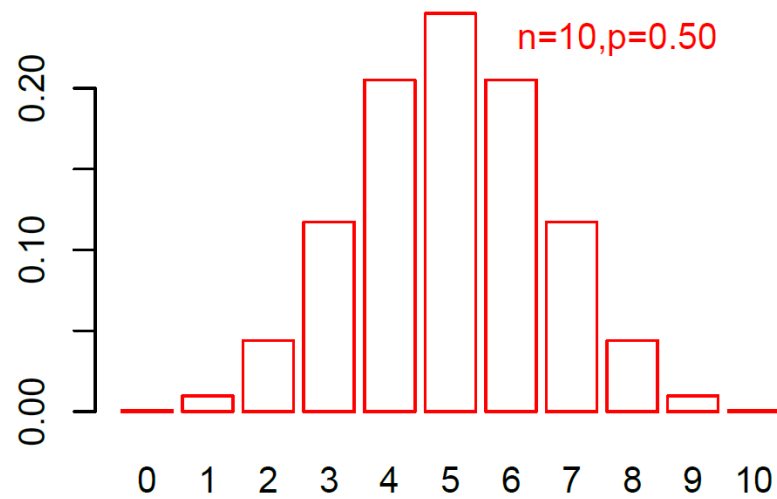
Derek Ogilvie

https://nl.wikipedia.org/wiki/Derek_Ogilvie



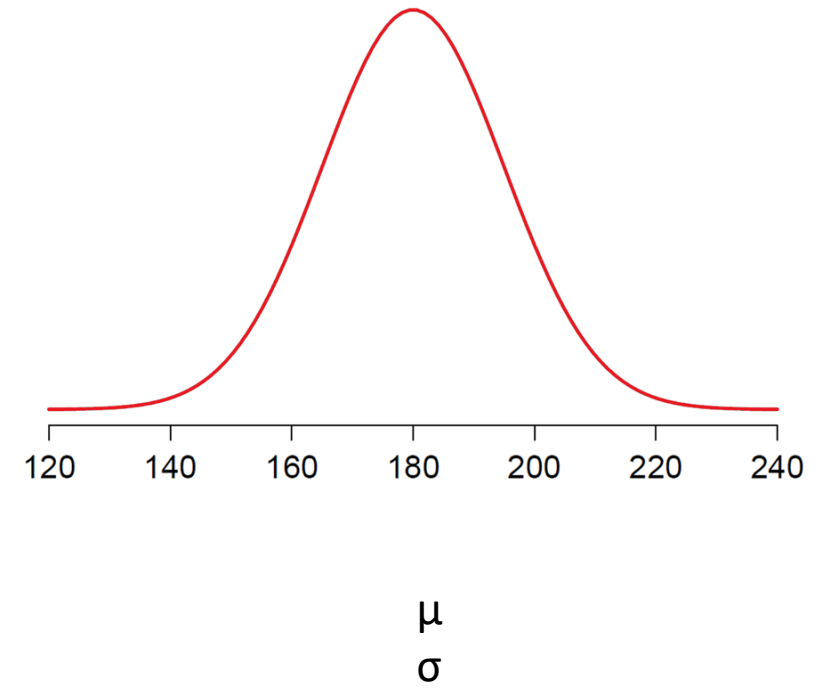
Thus:

Binomial distribution:



$$\mu = np$$
$$\sigma = \sqrt{np(1-p)}$$

Normal distribution:



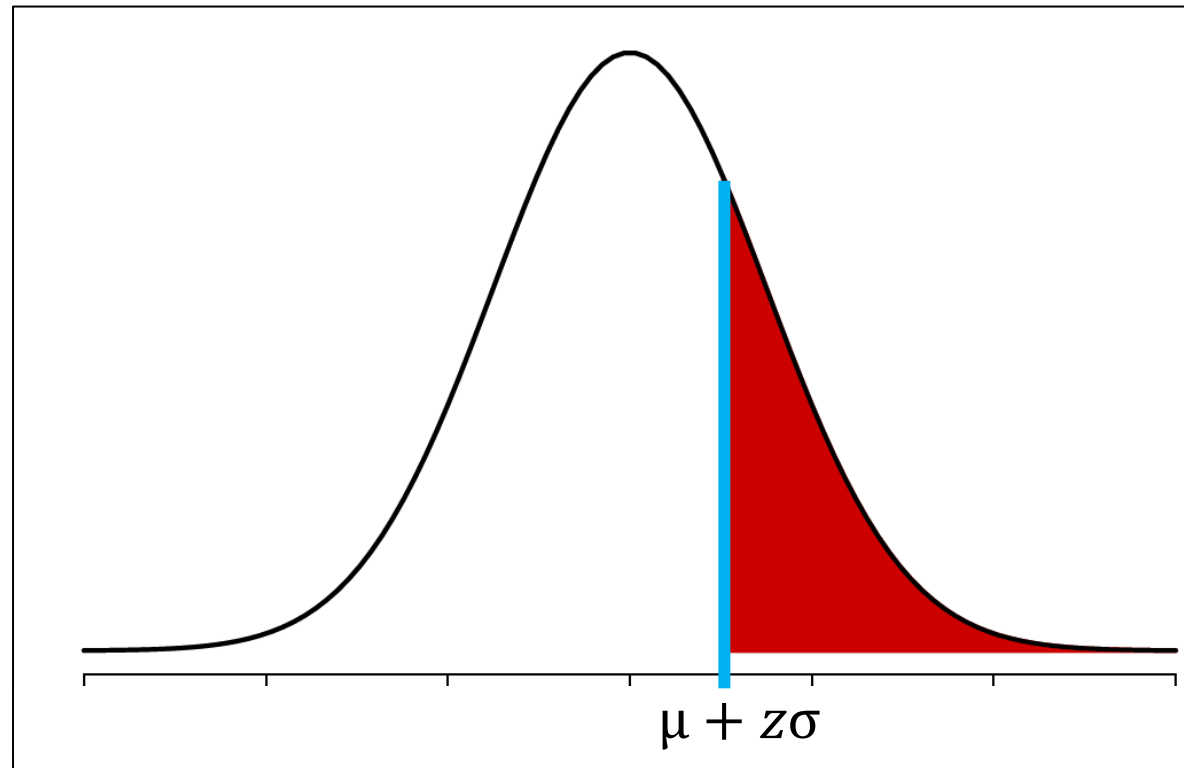
Today

- The normal distribution is a specific distribution that can hold for continuous variables. In case a continuous variable is normally distributed, you can use z-values to calculate probabilities of intervals of the distribution.
- The binomial random variable, X (number of successes in n independent trials), is an example of a discrete variable. The distribution of this variable is the binomial distribution.
- The binomial distribution is just like the normal distribution characterized by two parameters. For the normal distribution this is μ (the mean) and σ (the SD) and for the binomial distribution this is p (the probability of a success) and n (the number of trials).
- With sufficient trials ($np > 15$ and $n(1-p) > 15$) the binomial distribution will approximate a normal distribution, such that the tricks learned for the normal distribution can be applied there as well.

Practice exam question

Consider the normal distribution below, what is the z-score at the blue line given that the red area has a probability mass of 0.3?

- a) 0.5244
- b) 0.7580
- c) 0.6179



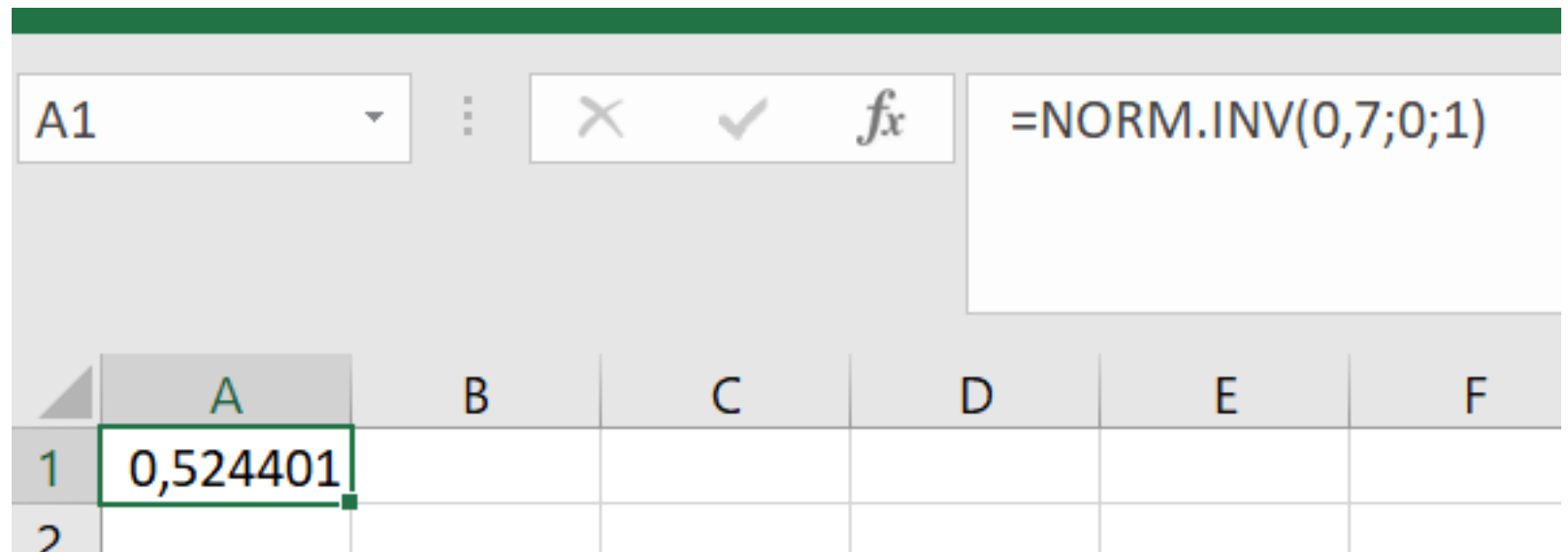
Answer

Step 1: The probability on the left side of the z-value is $1 - 0.3 = 0.7$

Step 2: use excel to find the z-value associated with this left-tail probability:

The answer:

$Z = 0.5244$



The image shows an Excel interface. At the top, a formula bar displays the formula `=NORM.INV(0,7;0;1)`. Below the formula bar, a spreadsheet grid is visible with columns labeled A through F and rows numbered 1 and 2. Cell A1 is selected and contains the value `0,524401`.

	A	B	C	D	E	F
1	0,524401					
2						